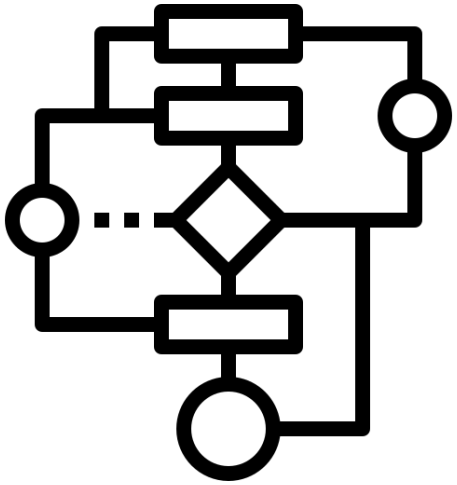
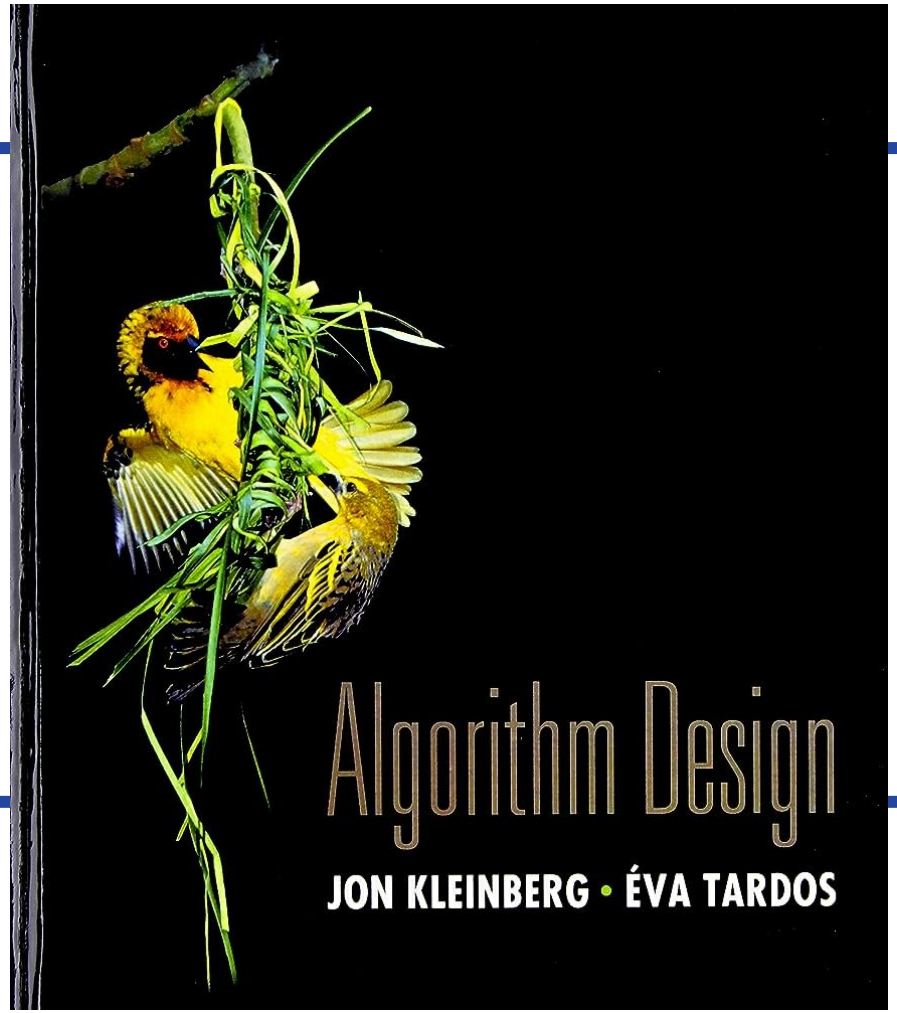


CS 310: Algorithms

Lecture 15

Instructor: Naveed Anwar Bhatti



Chapter 4:
Greedy Algorithms

Section 4.4: Shortest Path Algorithm
Dijkstra Algorithm

Administrivia

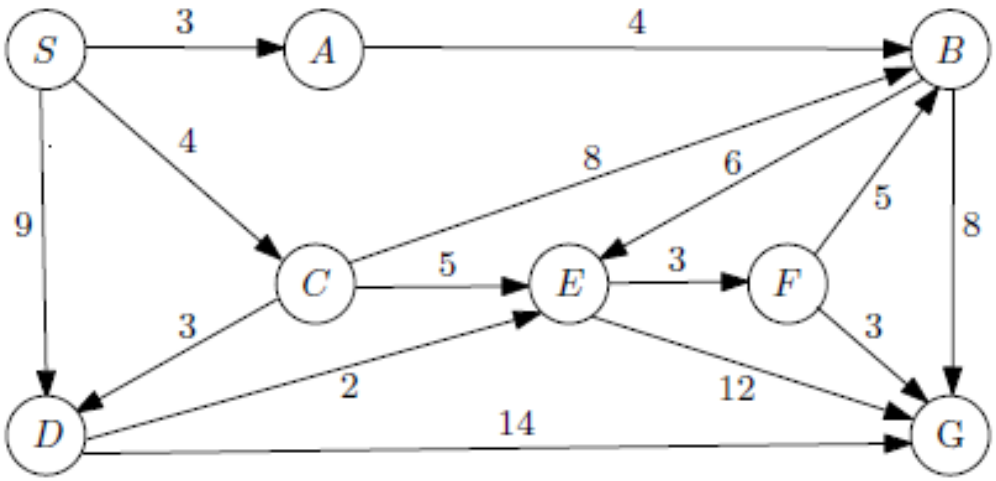
- Midterm marking not complete but **scene bad hai**
- Expect **next three quizzes** on following dates:

Quiz 4	Quiz 5	Quiz 6
6th November	20th November	4th December

- **Assignment 3** will be released soon

Weighted (Directed) Graph

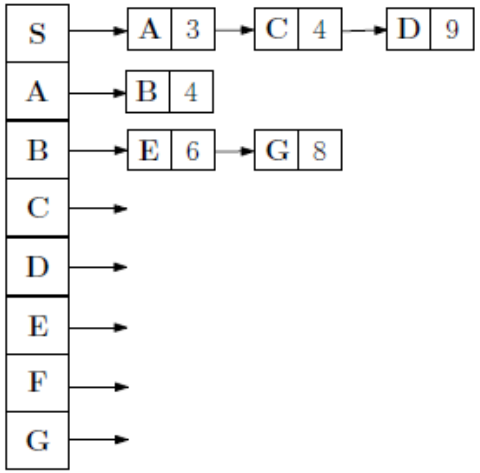
- V : Set of vertices
- E : Set of edges (directed edges)
- w : cost/weight function: $w : E \to \mathbb{R}$
- weights could be lengths, airfare, toll, energy
- Denoted by $G = (V, E, w)$



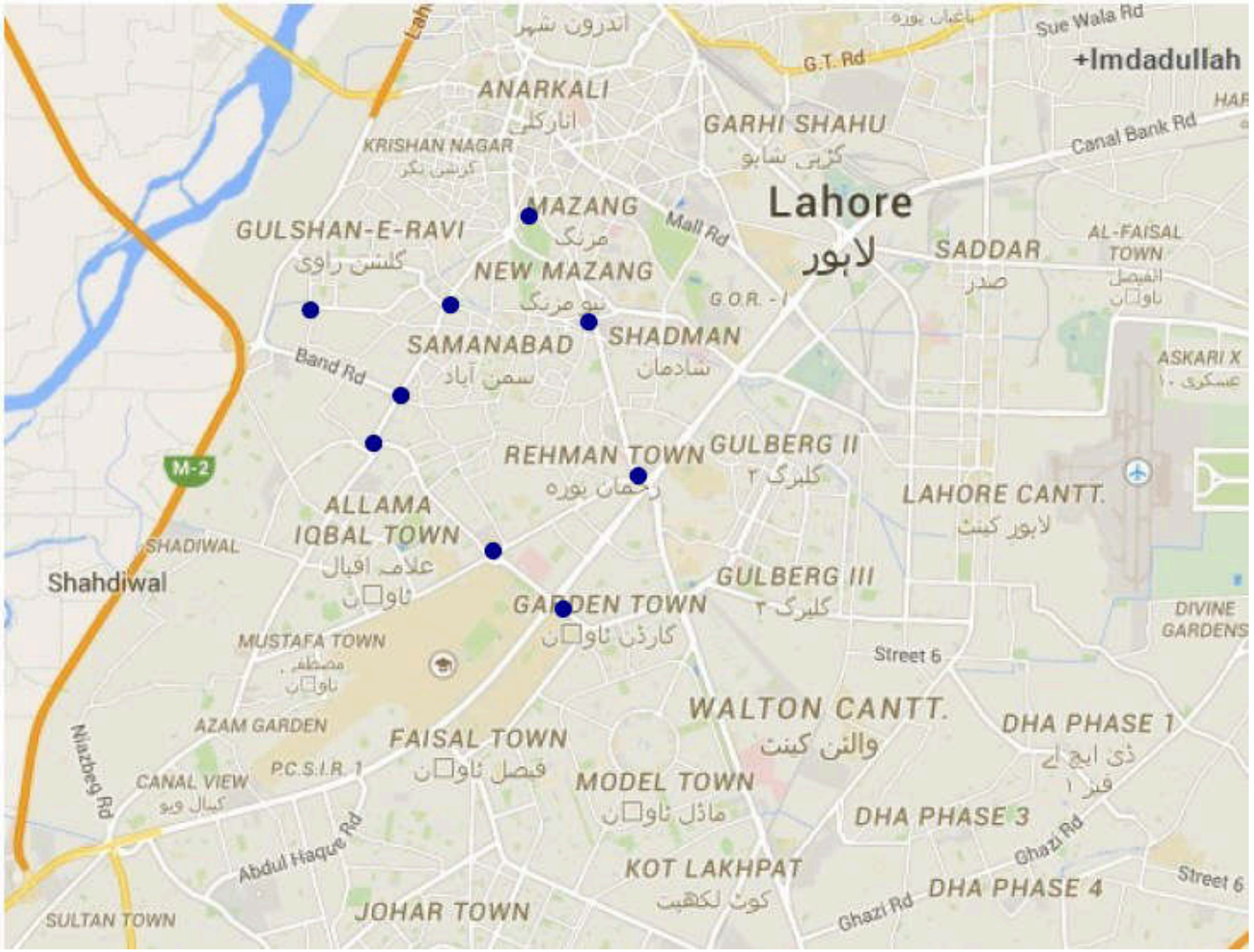
Weighted Adjacency Matrix

	S	A	B	C	D	E	F	G
S	0	3	0	4	9	0	0	0
A	0	0	4	0	0	0	0	0
B	0	0	0	0	0	6	0	8
C	⋮							
D								
E								
F								
G								

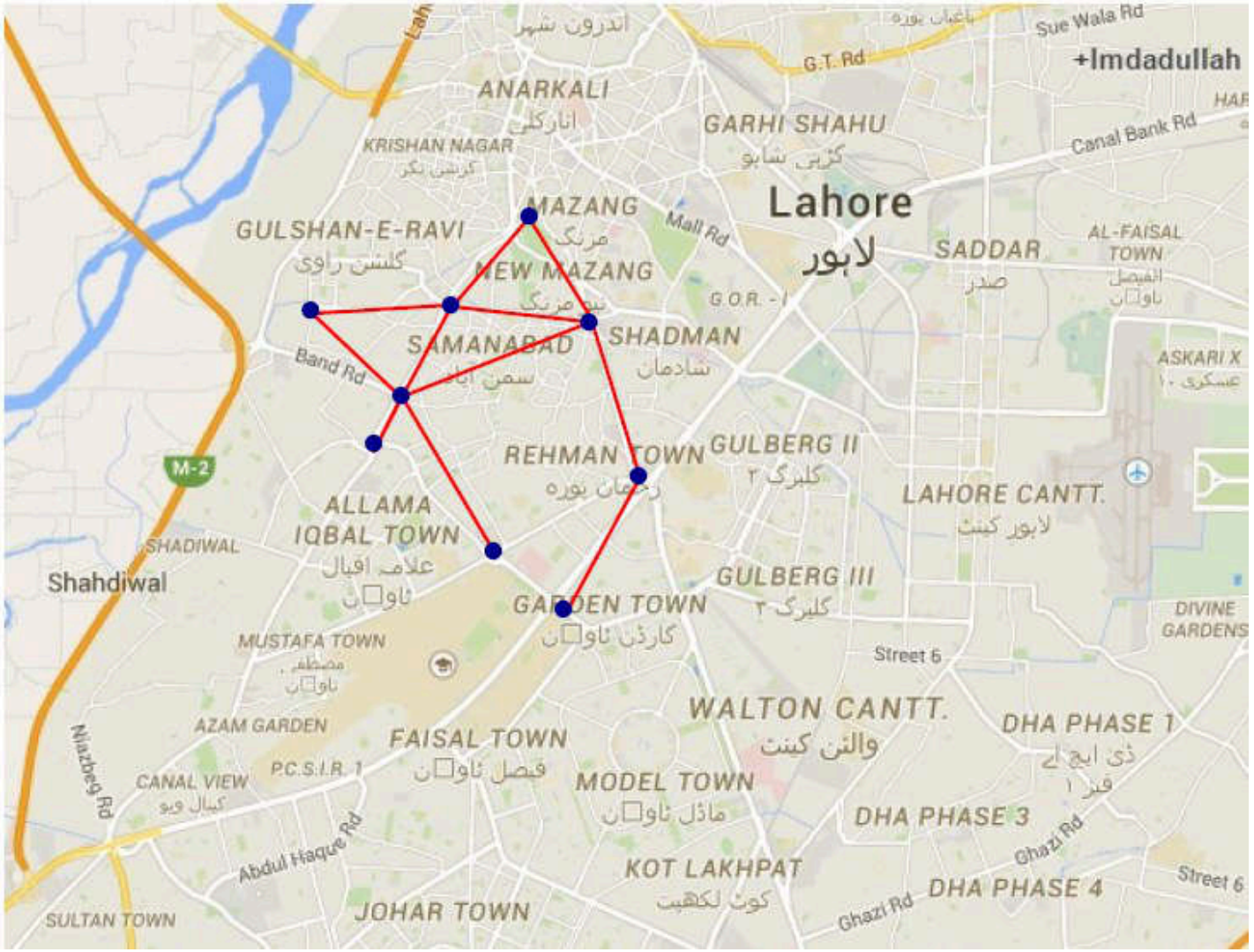
Weighted Adjacency Lists



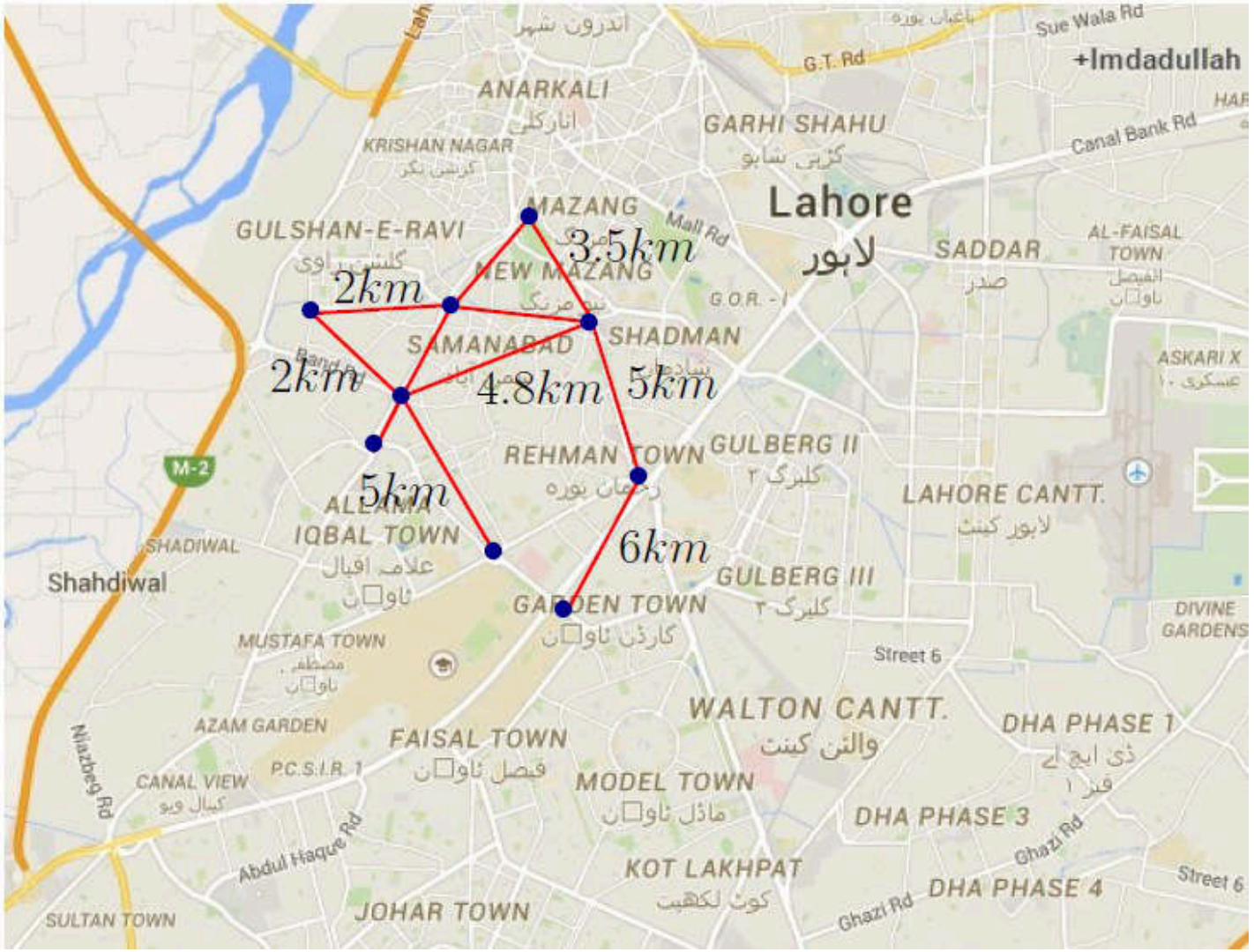
Weighted (Directed) Graph



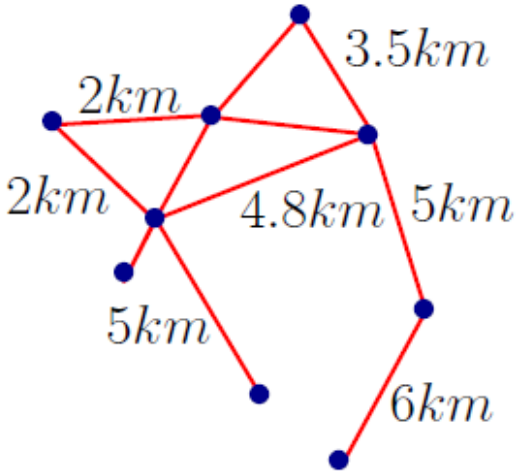
Weighted (Directed) Graph



Weighted (Directed) Graph

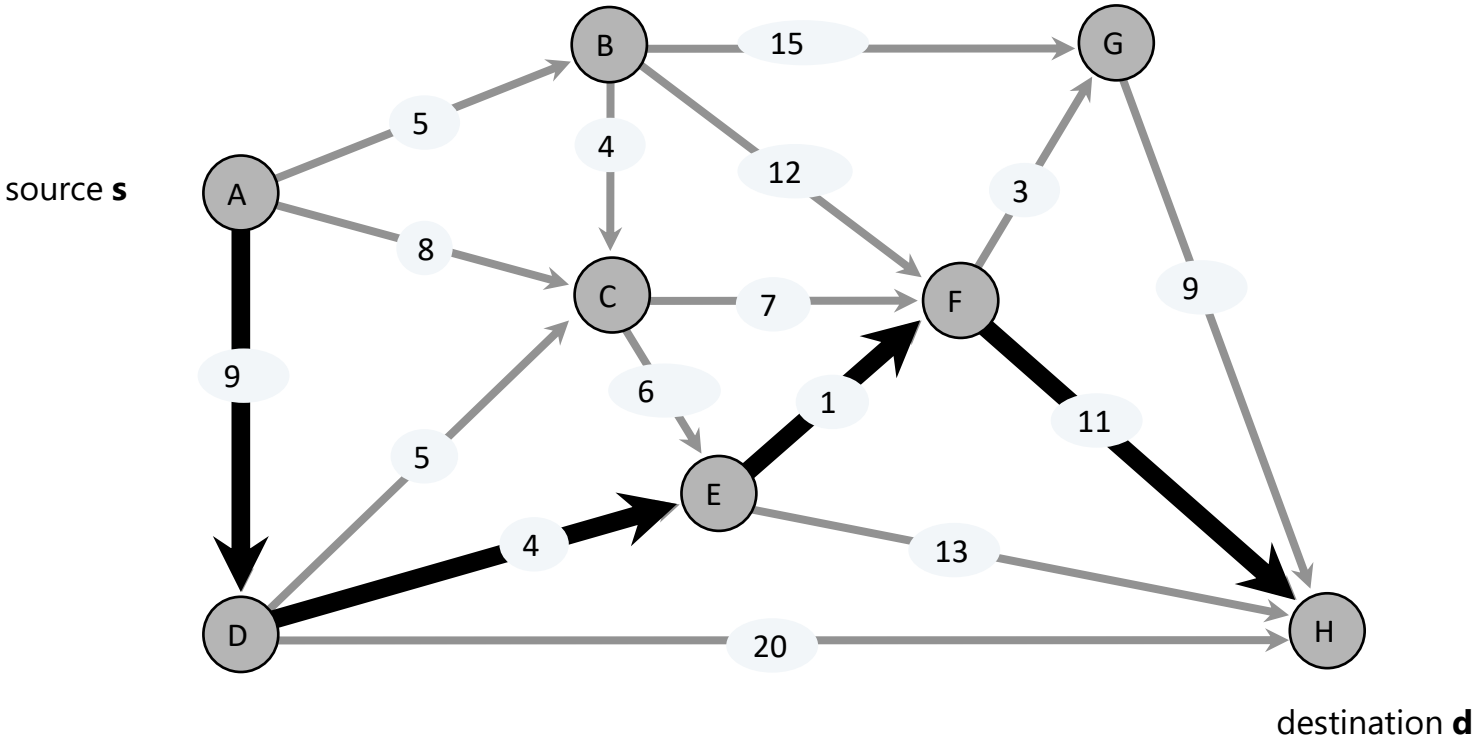


Weighted (Directed) Graph



Single-pair shortest path problem

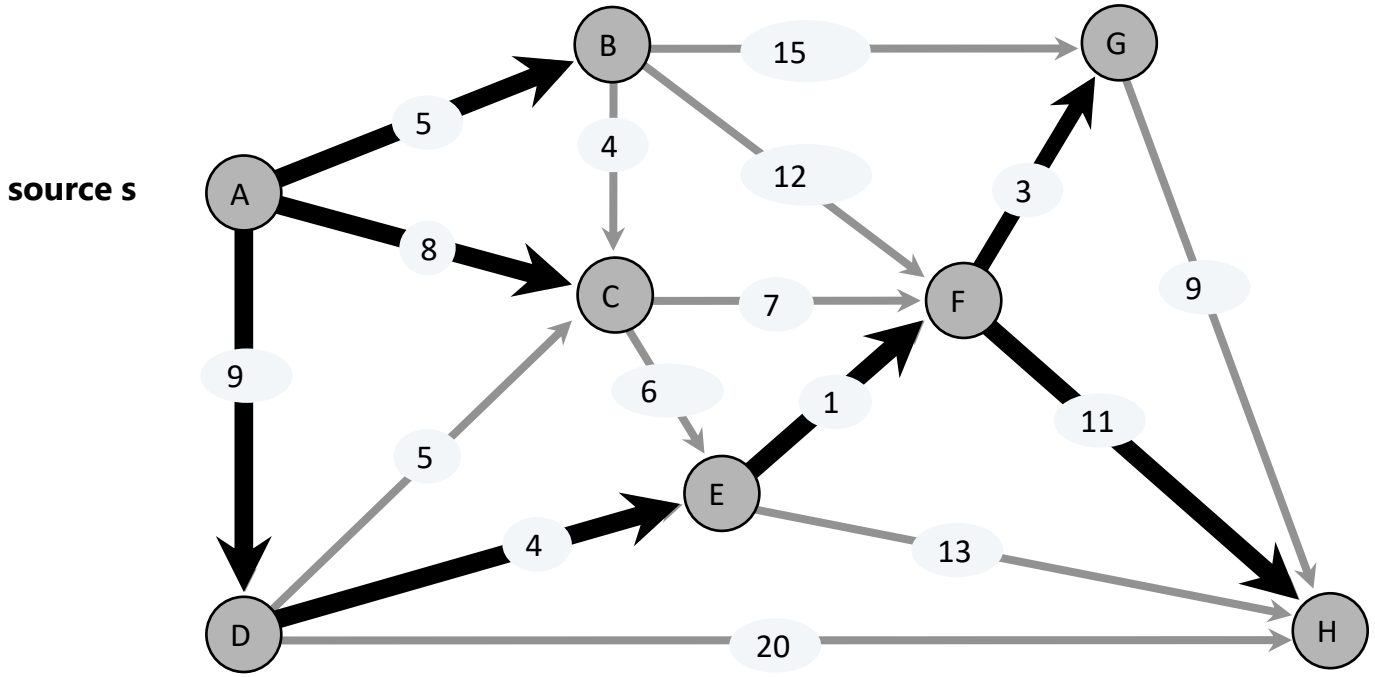
- **Problem:** Given a digraph $G = (V, E, w)$, source $s \in V$, and destination $d \in V$, find a shortest directed path from s to d.



length of path = 9 + 4 + 1 + 11 = 25

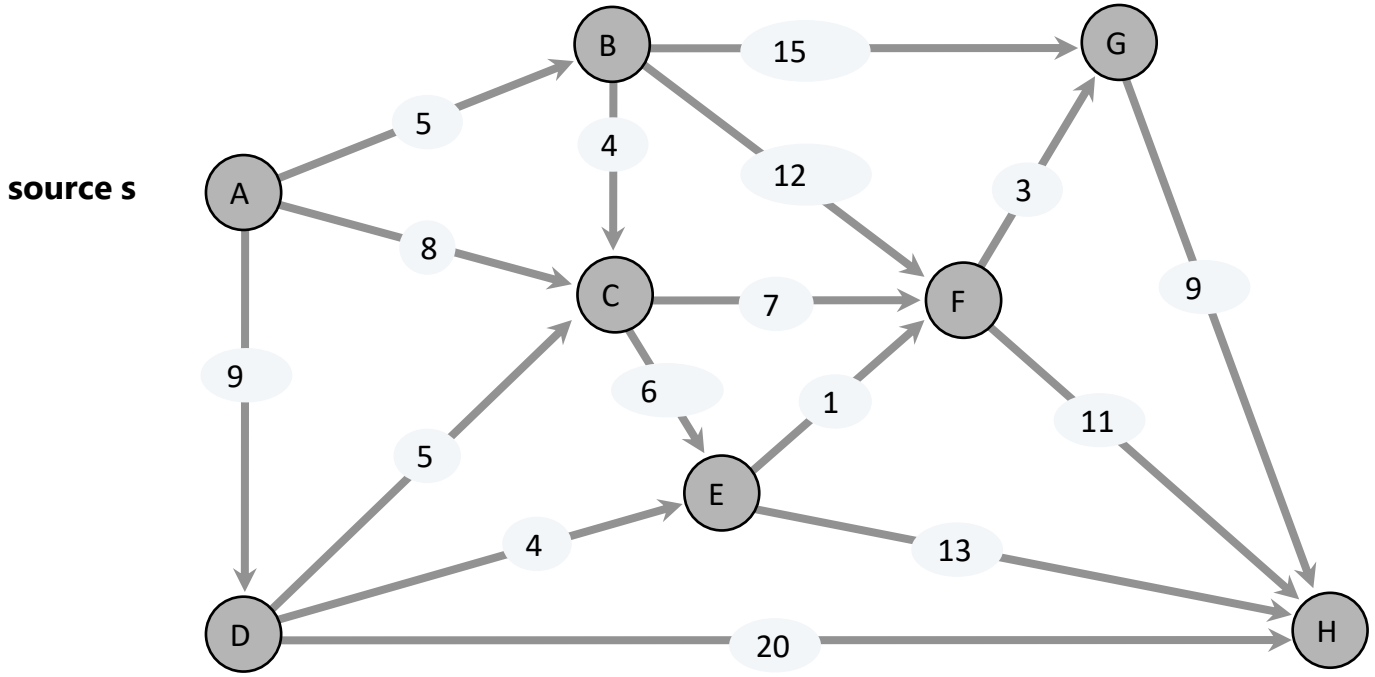
Single-source shortest paths problem

- **Problem**: Given a digraph $G = (V, E, w)$, source $s \in V$, find a shortest directed path from s to every node.



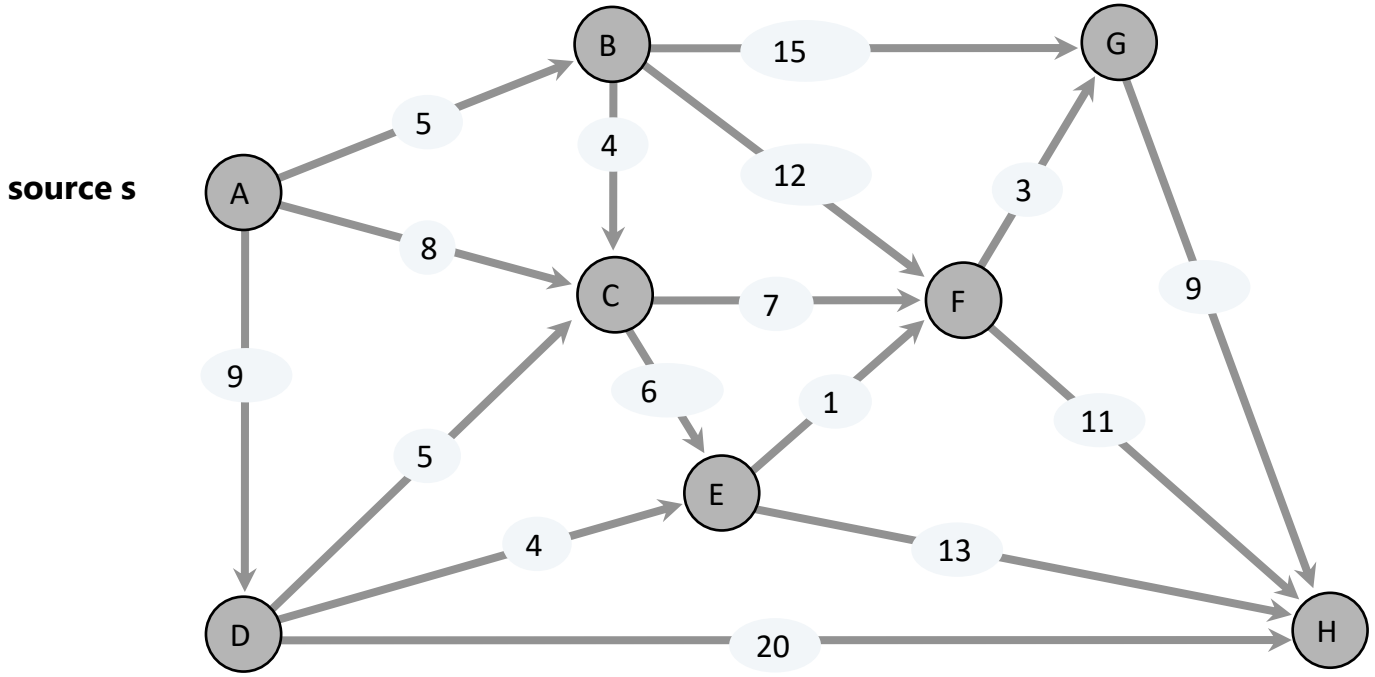
shortest-paths tree

Single-source shortest paths problem



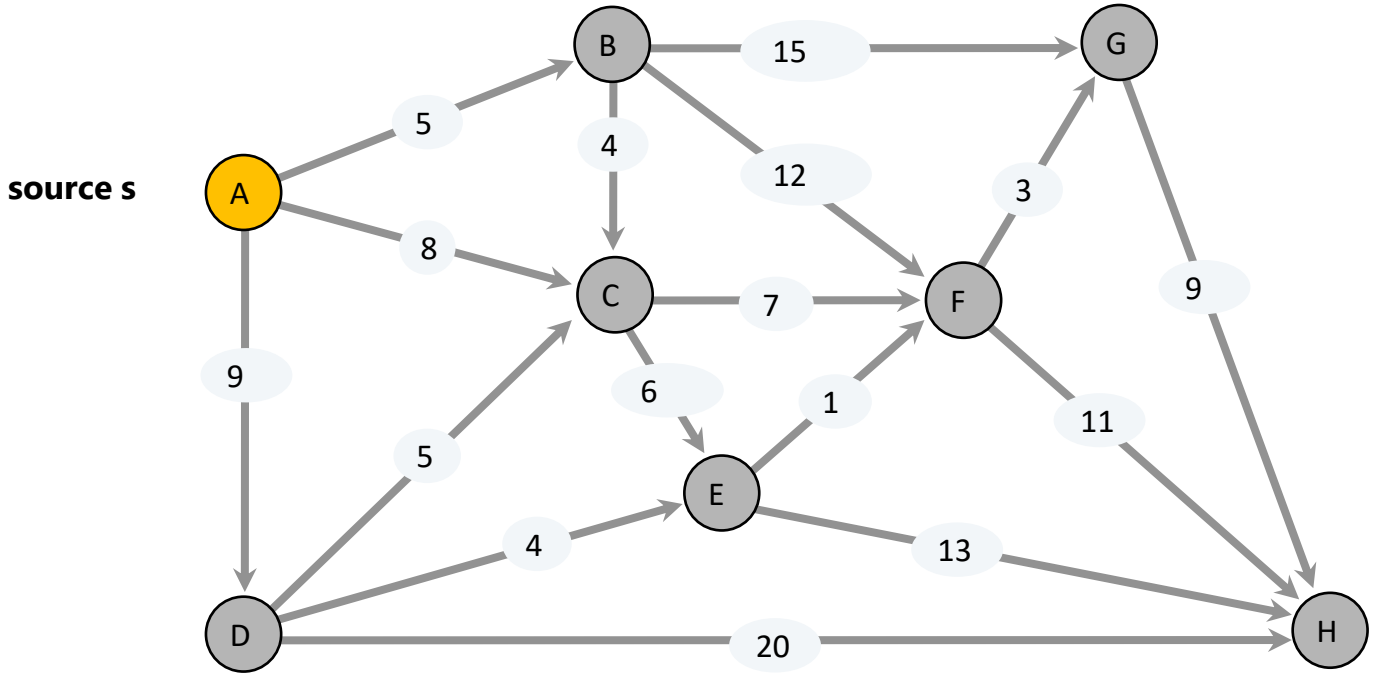
Single-source shortest paths problem

Why can't we use BFS?



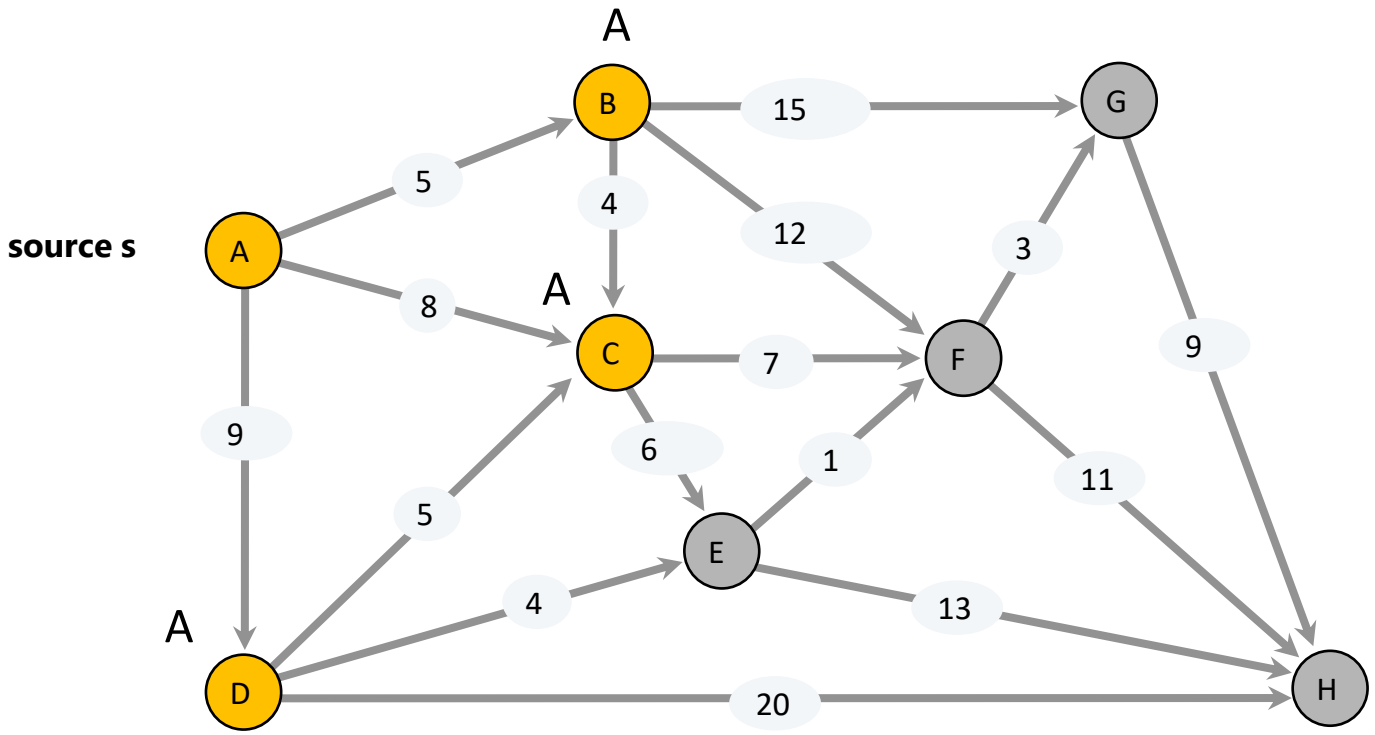
Single-source shortest paths problem

Why can't we use BFS?



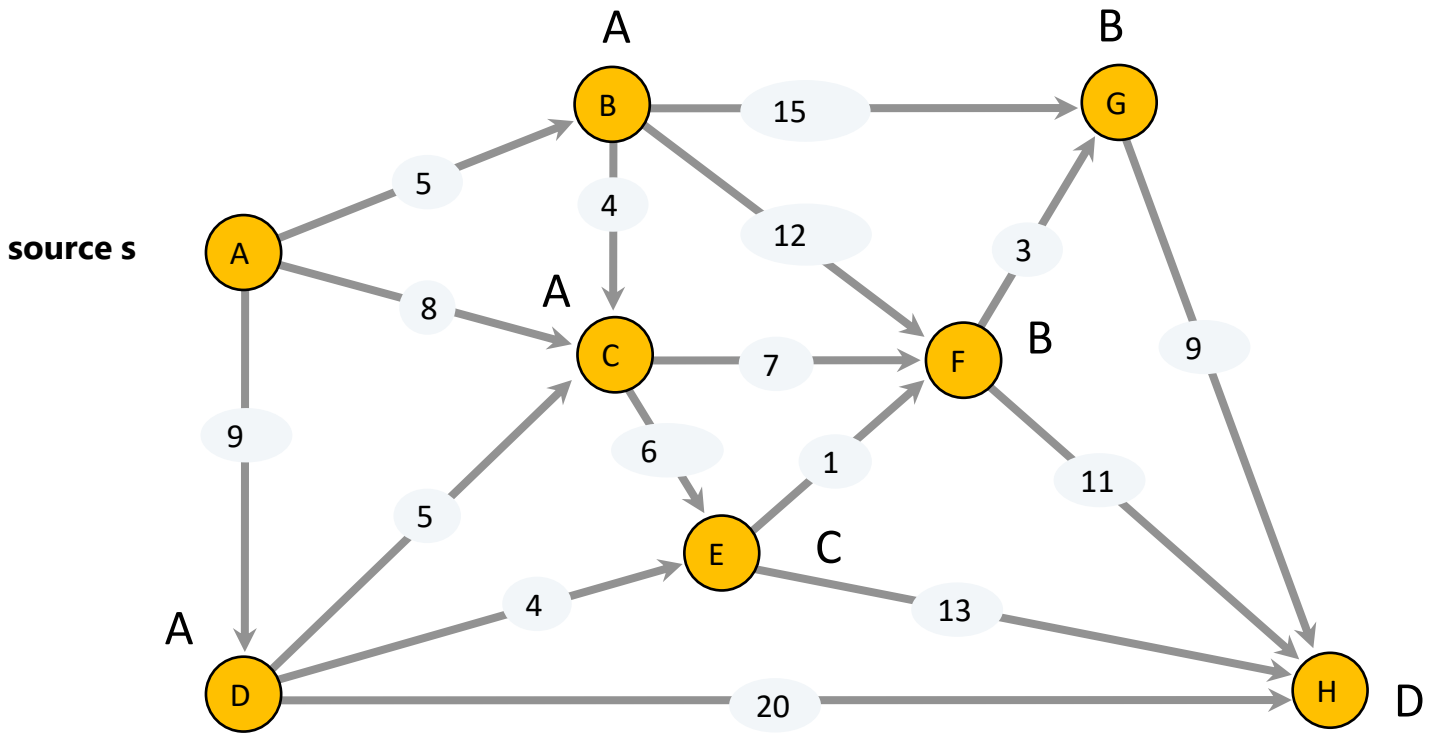
Single-source shortest paths problem

Why can't we use BFS?



Single-source shortest paths problem

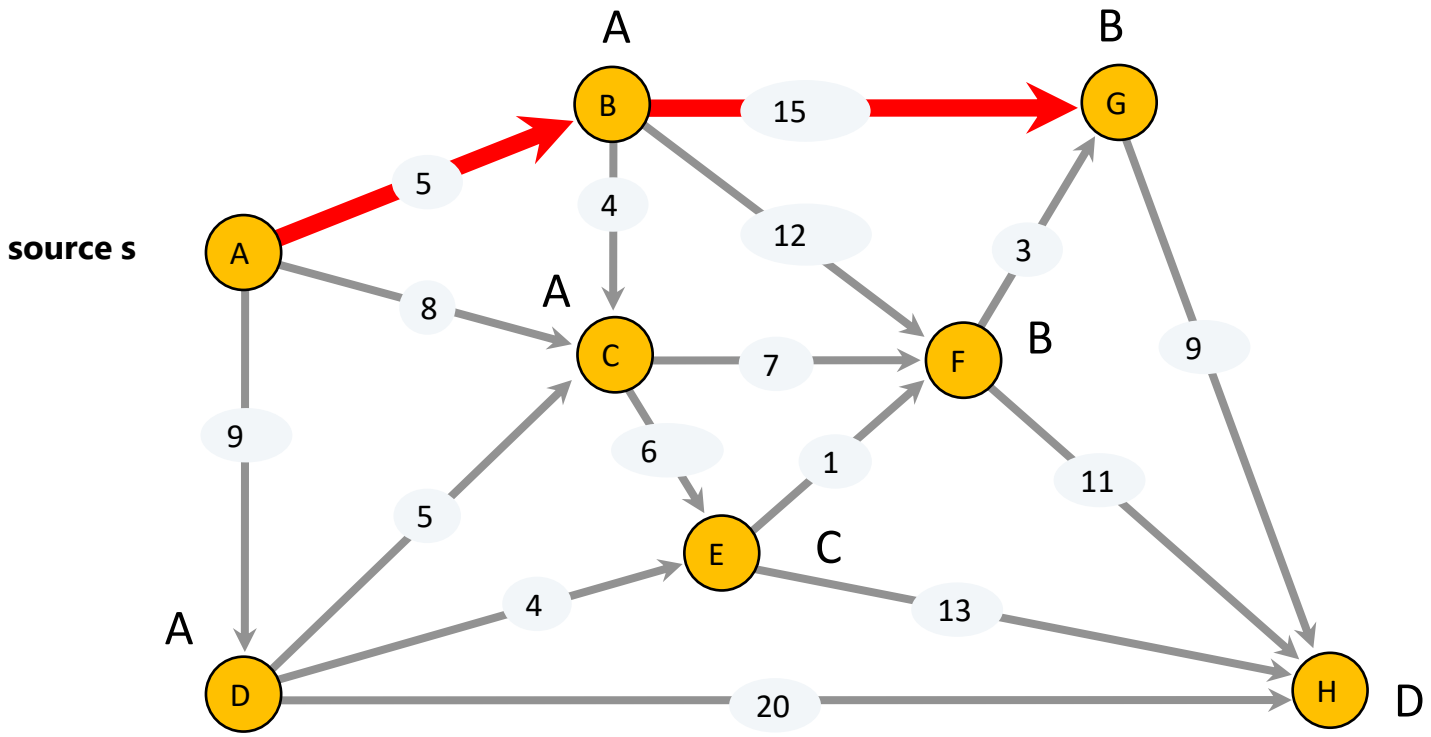
Why can't we use BFS?



Single-source shortest paths problem

Why can't we use BFS?

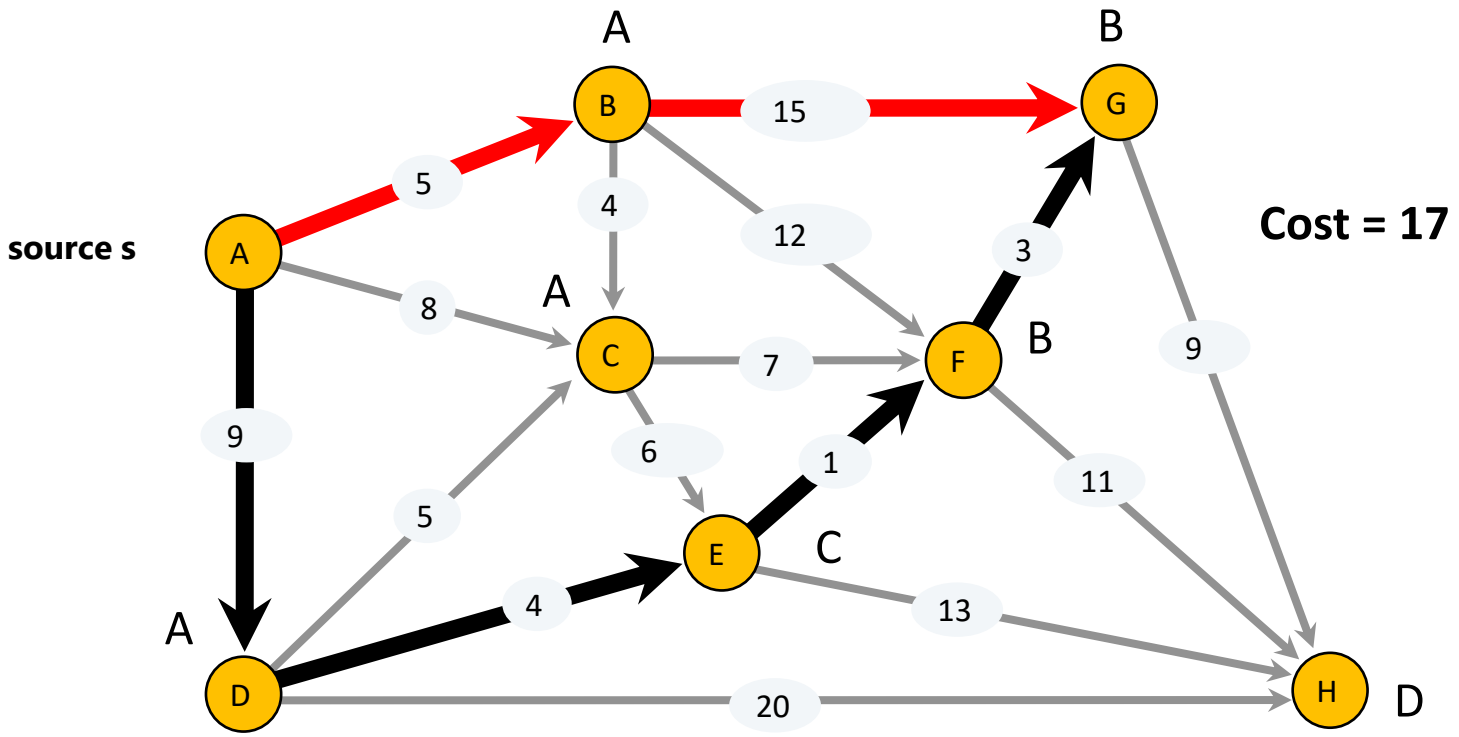
Cost = 20



Single-source shortest paths problem

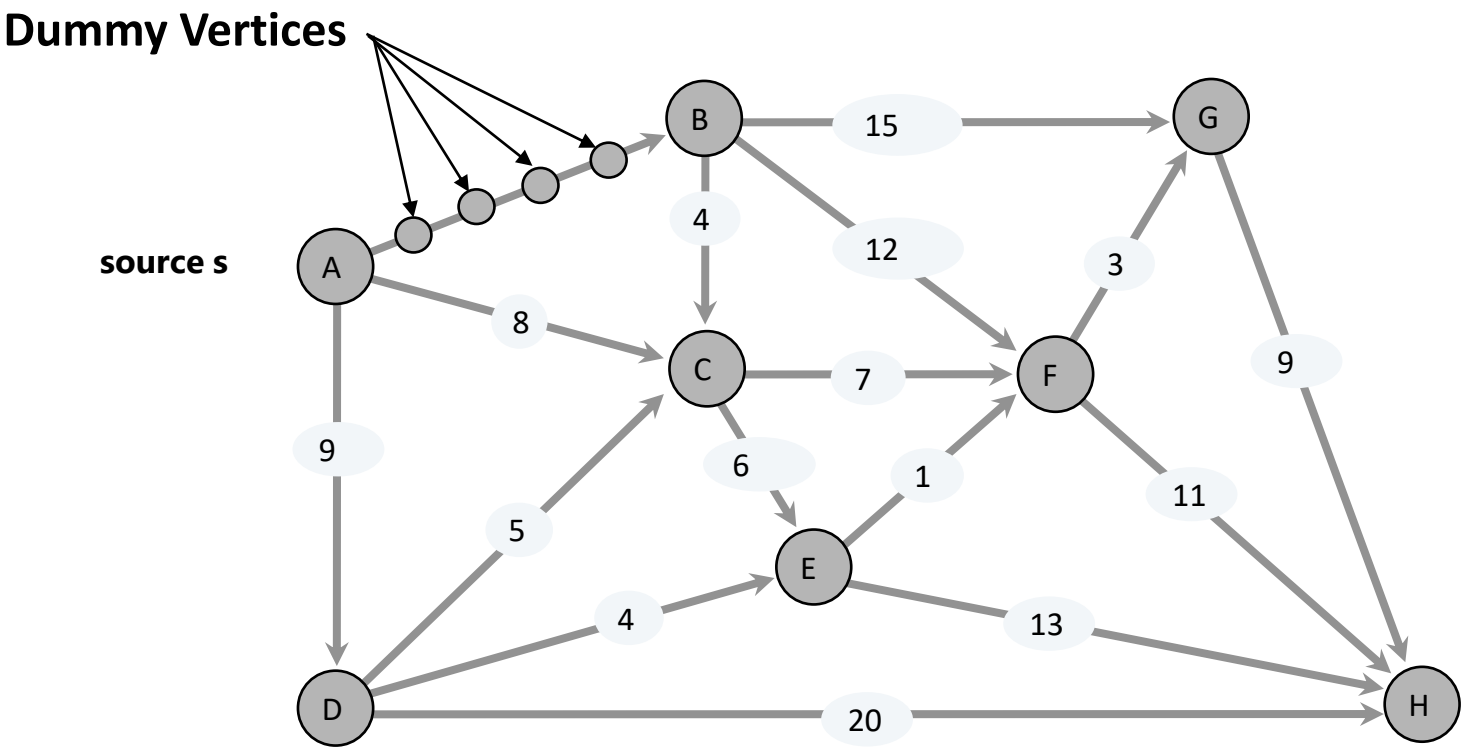
Why can't we use BFS?

Cost = 20



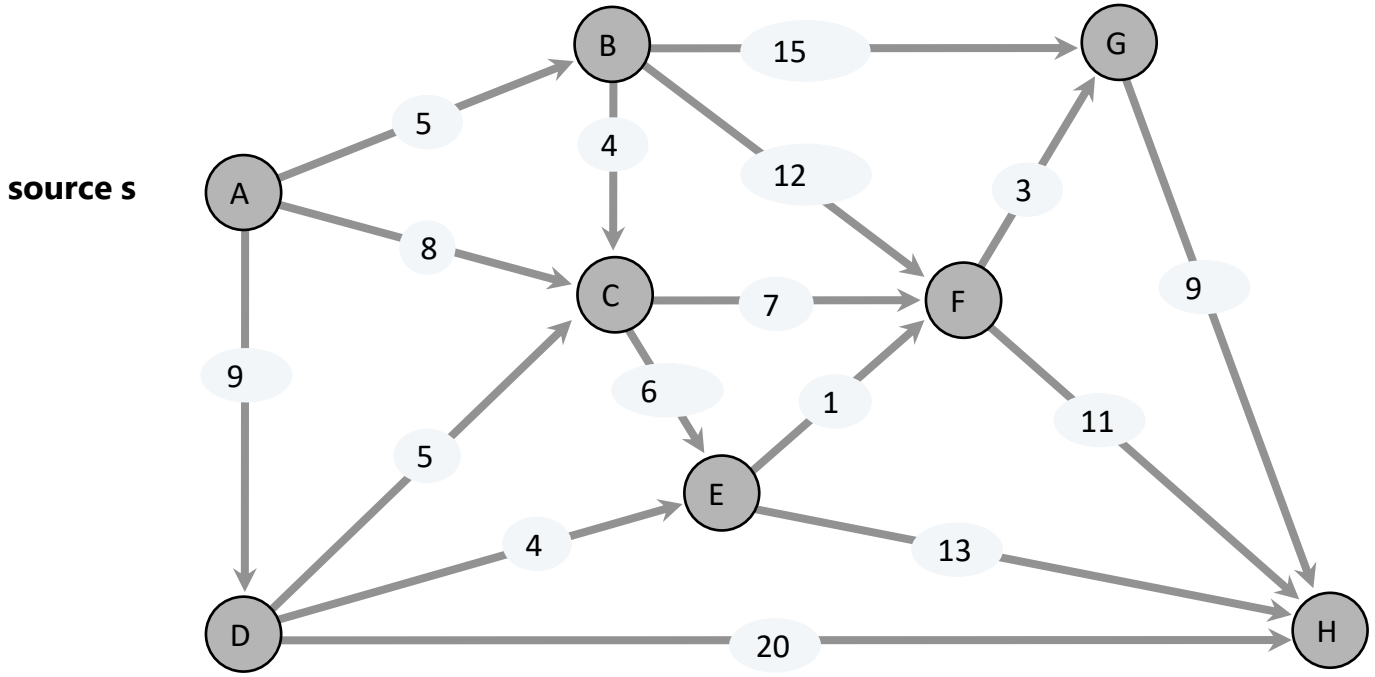
Cost = 17

Single-source shortest paths problem



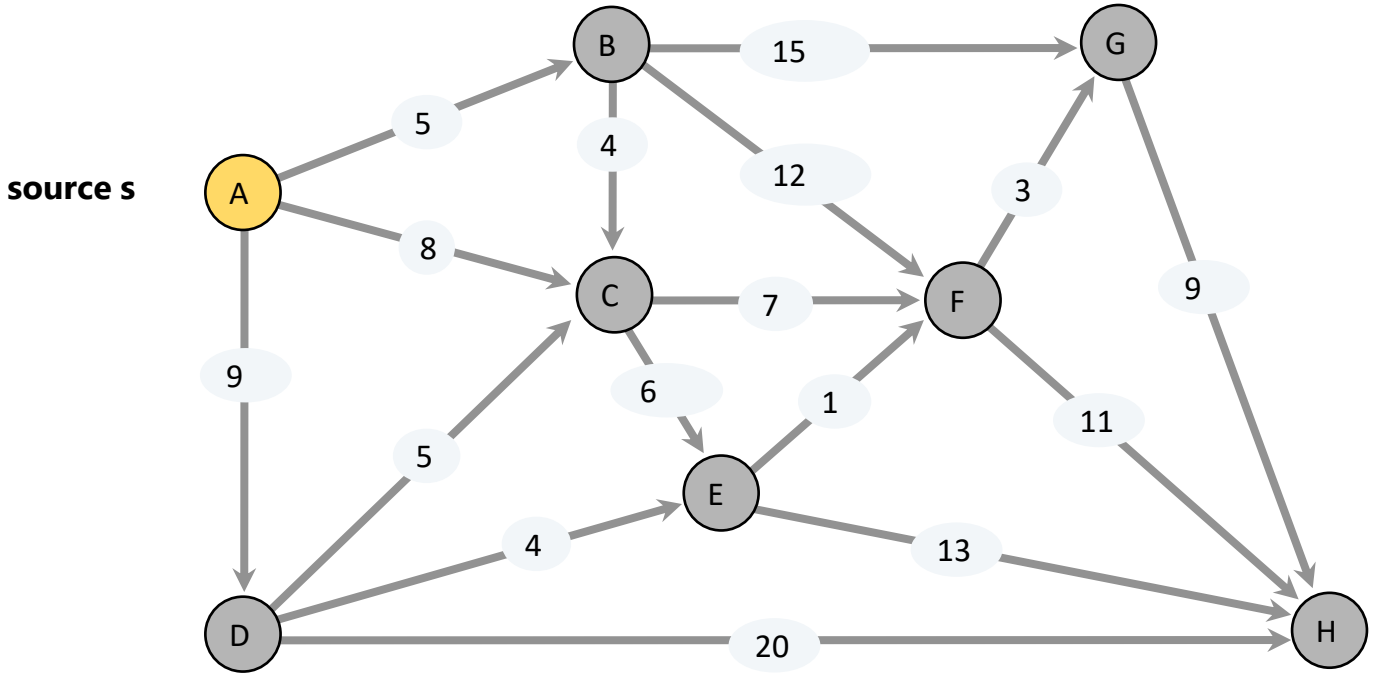
Edge splitting: Transforming a weighted graph into an unweighted one

Single-source shortest paths problem



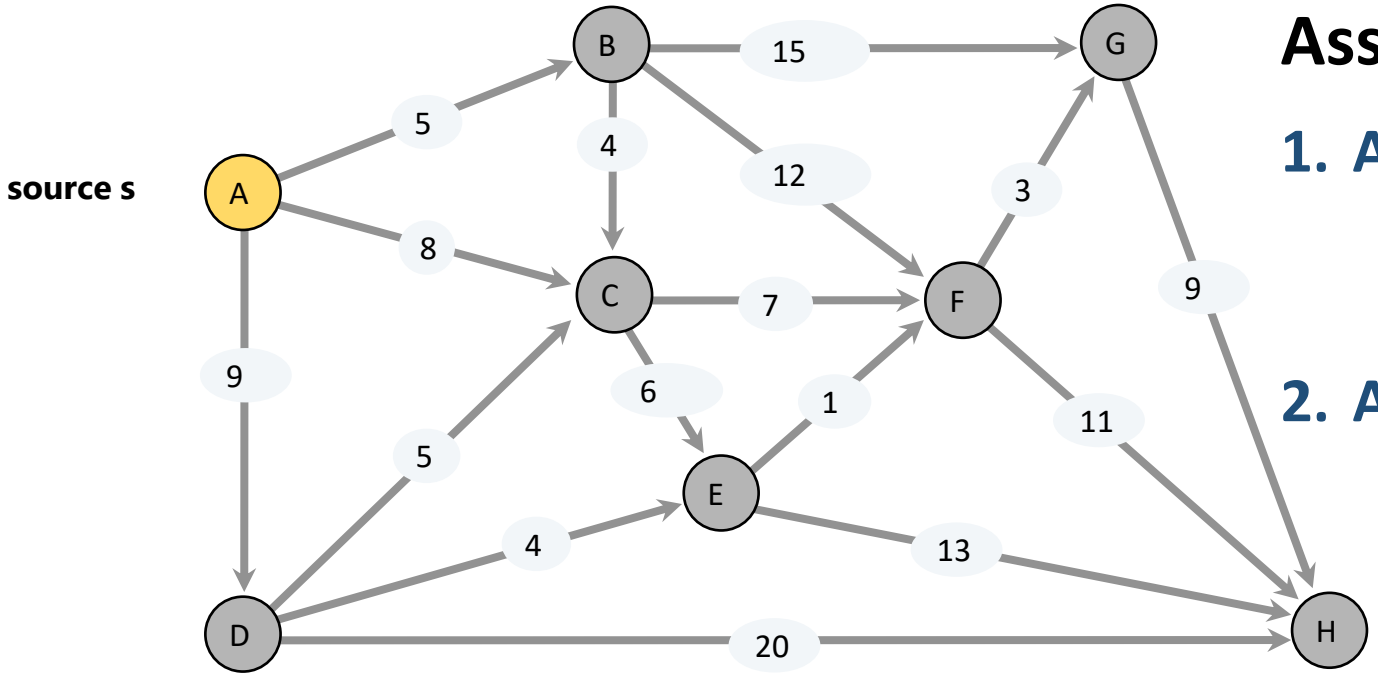
Dijkstra's algorithm for shortest path

Single-source shortest paths problem



Dijkstra's algorithm solves this problem for both **directed** and **undirected** graphs

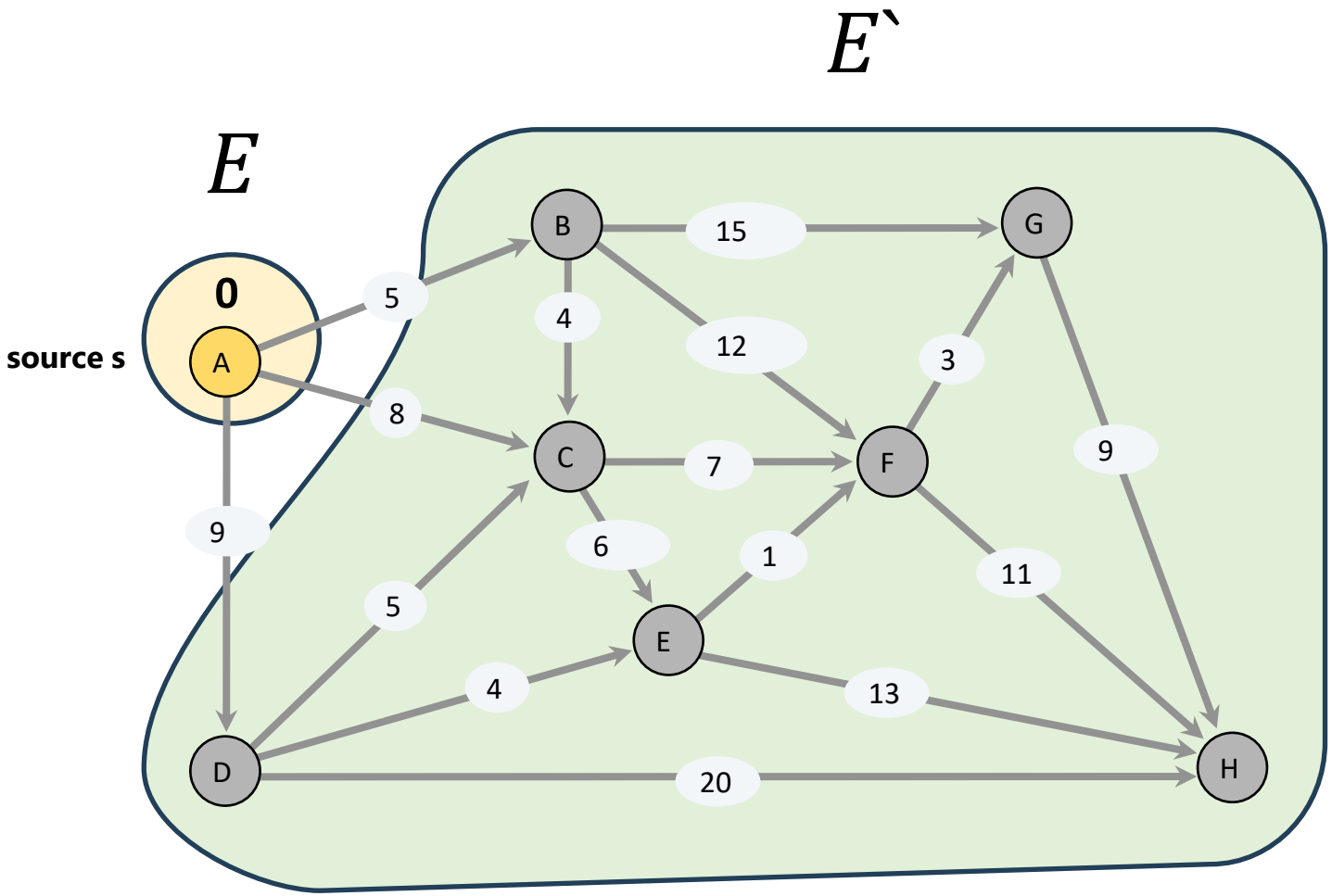
Single-source shortest paths problem



Assumption:

1. **All vertices are reachable from s**
   - Otherwise, there is no shortest path
   - Easy to get R(s) in preprocessing (e.g., BFS or DFS)
2. **All edge weights are non-negative**

Single-source shortest paths problem

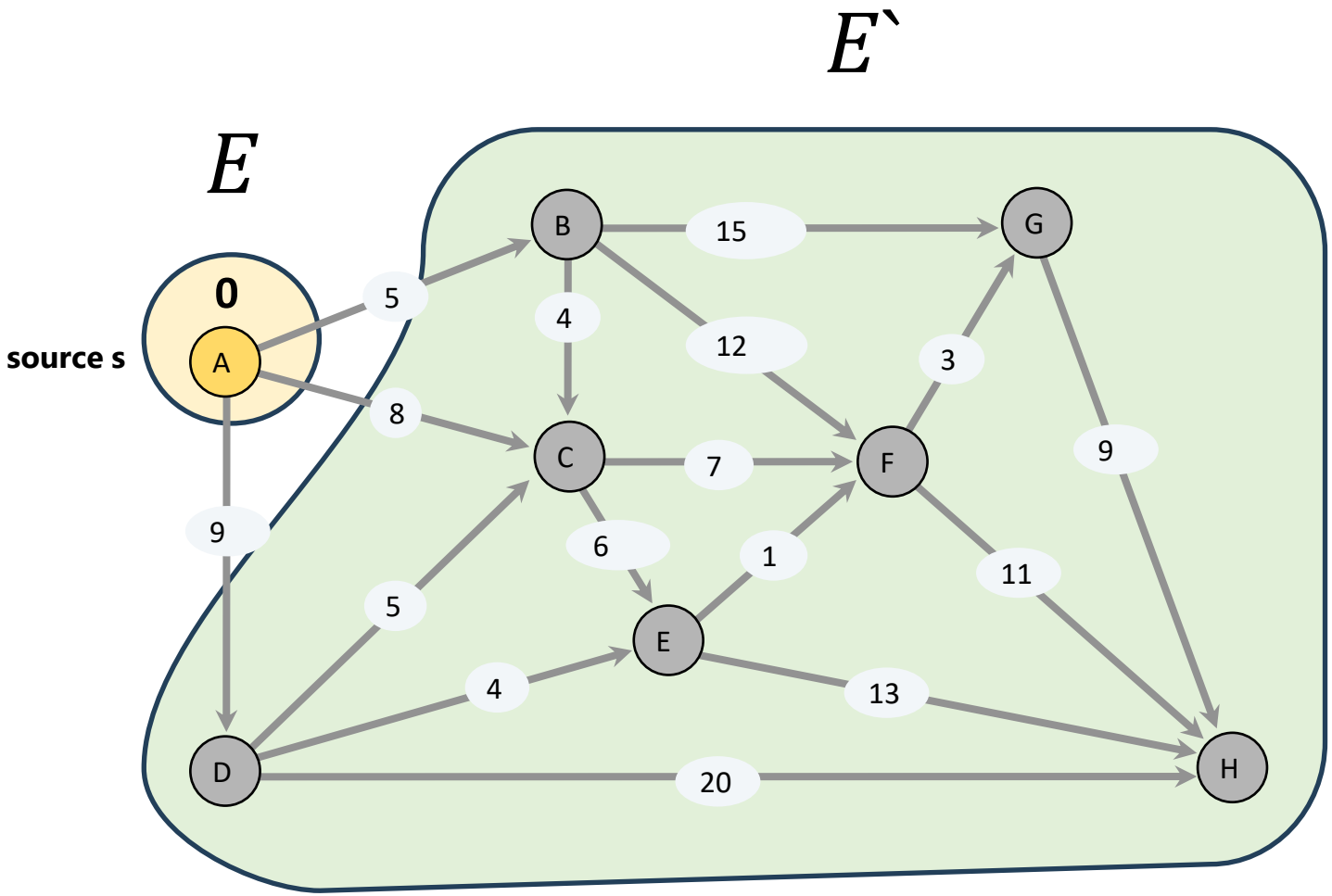


distance[ ]	
A	**0**
B	∞
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

prev[ ]	
A	
B	
C	
D	
E	
F	
G	
H	

Initially ***E*** = {A} and iteratively add one vertex to ***E`***

Single-source shortest paths problem

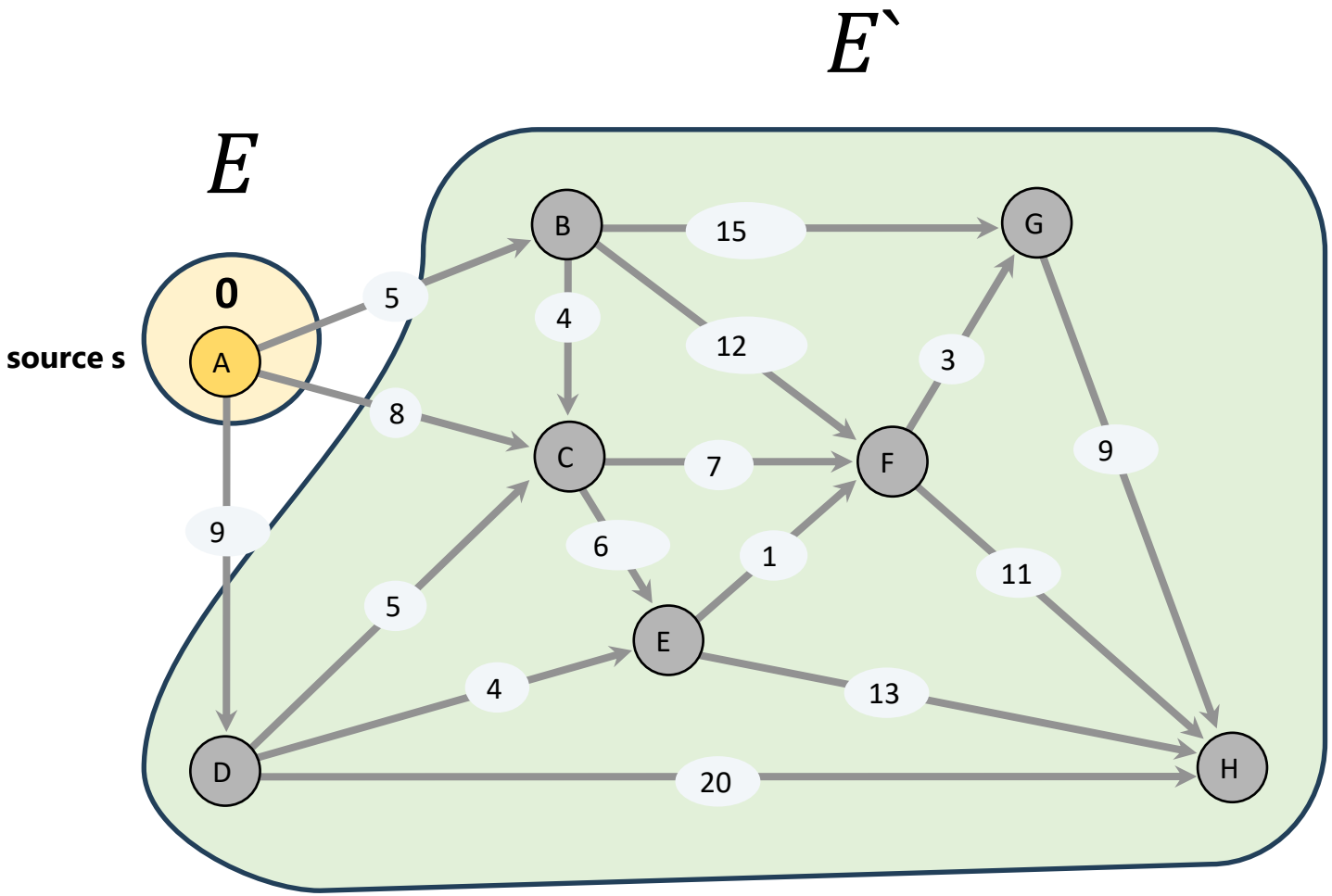


distance[ ]	
A	**0**
B	∞
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

prev[ ]	
A	
B	
C	
D	
E	
F	
G	
H	

Which vertex from **E`** to add to **E**?

Single-source shortest paths problem

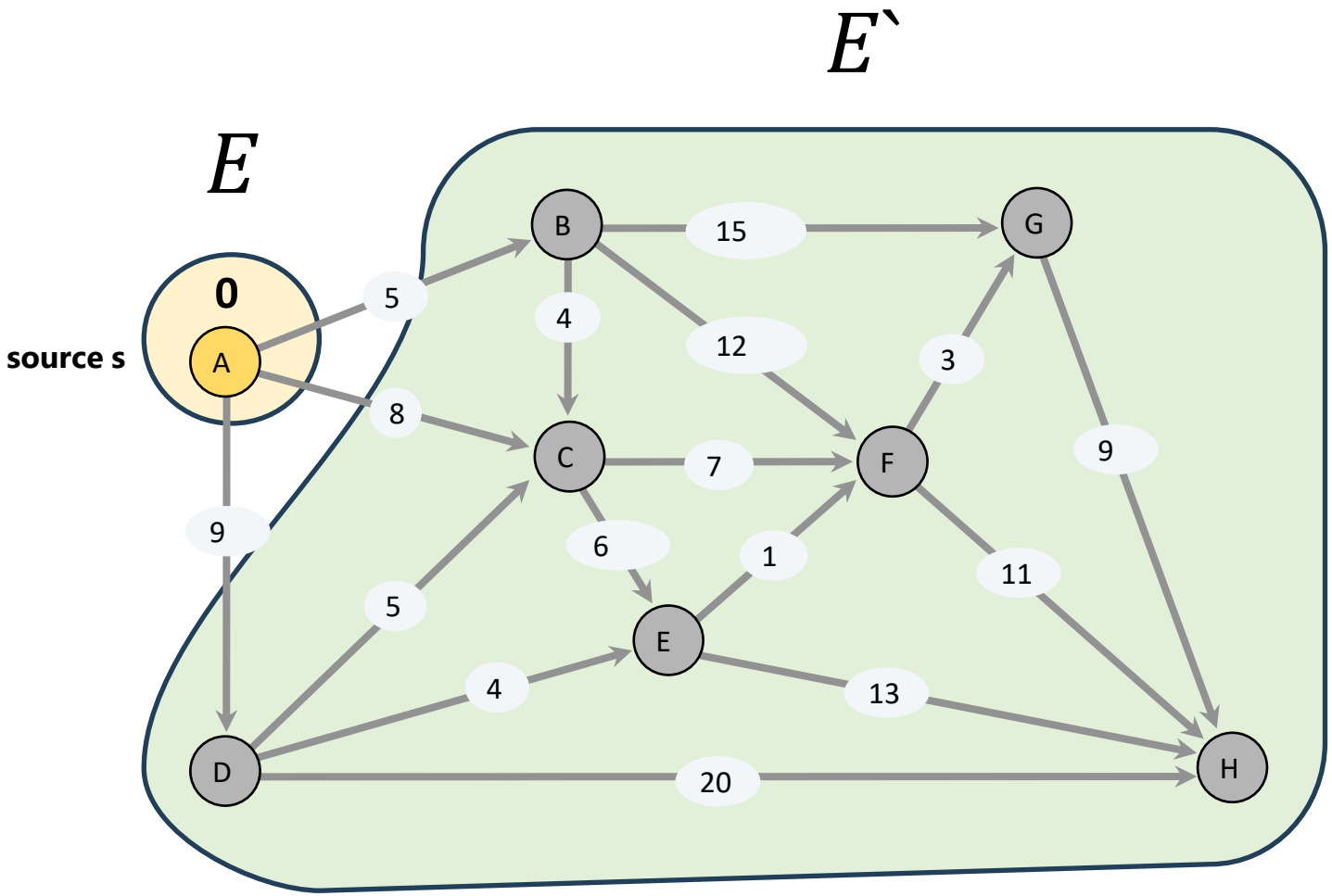


distance[ ]	
A	**0**
B	∞
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

prev[ ]	
A	
B	
C	
D	
E	
F	
G	
H	

The vertex $v \epsilon E`$ that is closest to **A**

Single-source shortest paths problem



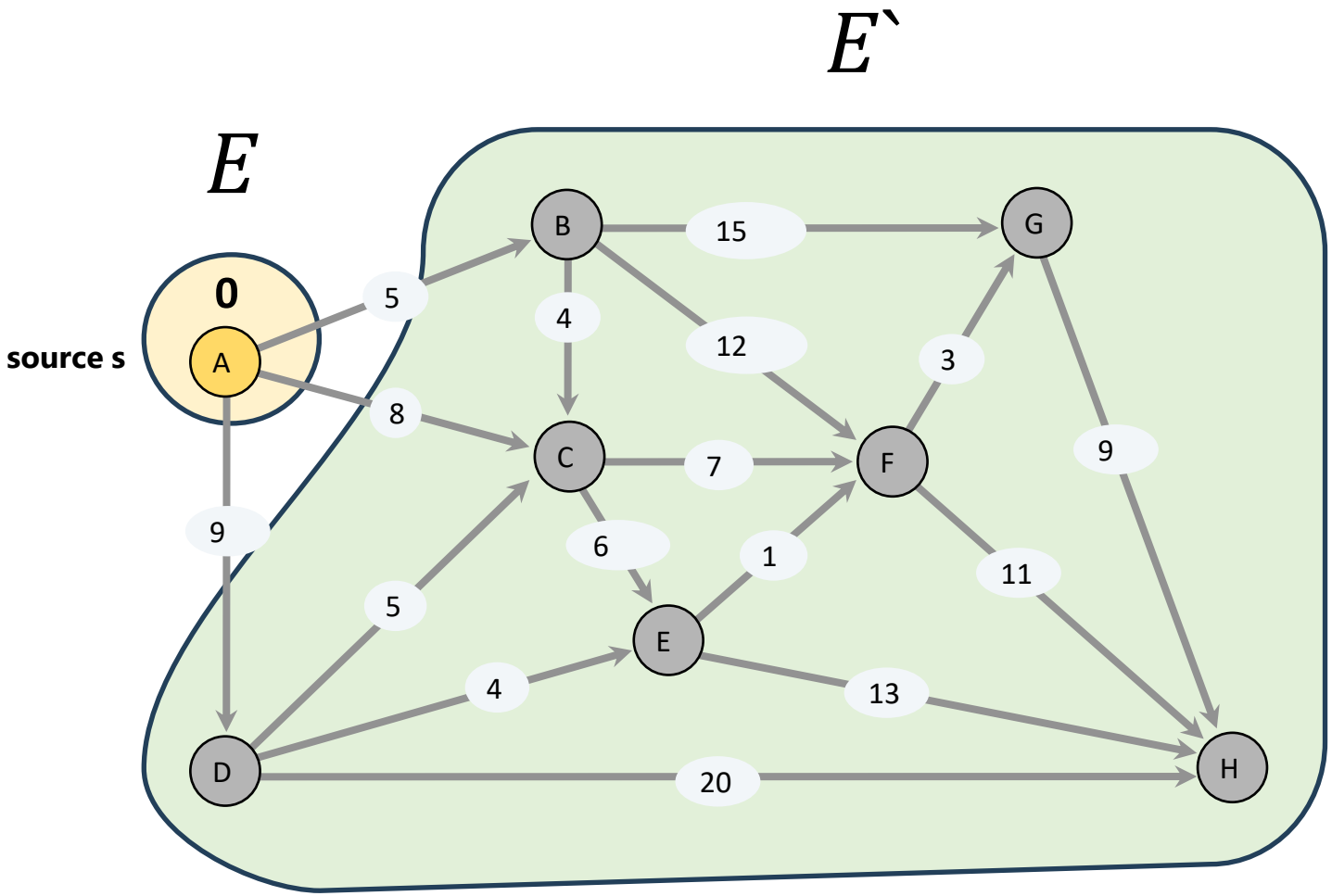
distance[ ]	
A	0
B	∞
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

prev[ ]	
A	
B	
C	
D	
E	
F	
G	
H	

$$\pi(v) = \min_{e=(u,v)\,:\,u \in S} \boxed{d[u] + \ell_e}$$

the length of a shortest path from s to some node u in explored part E, followed by a single edge $e = (u, v)$

Single-source shortest paths problem

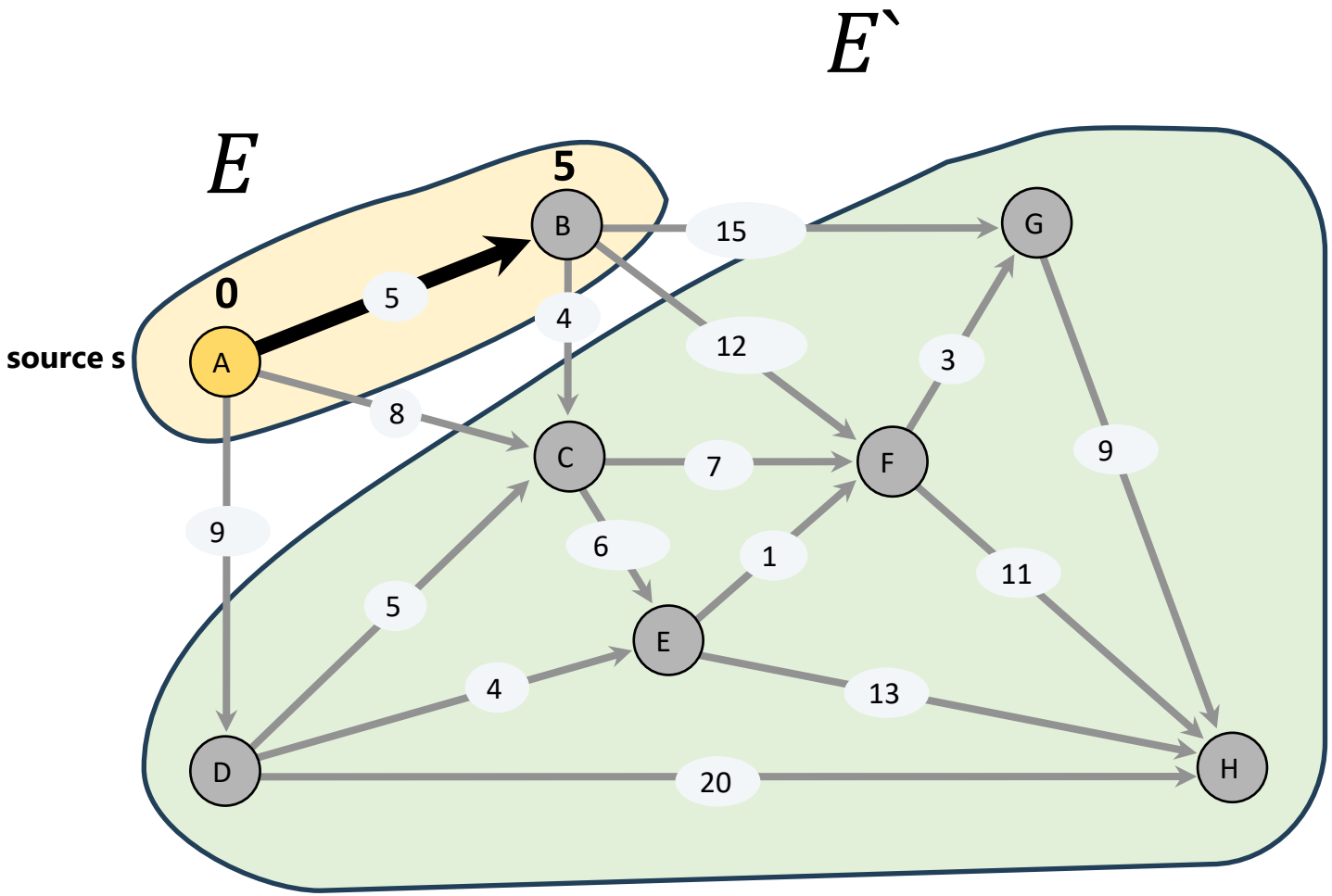


distance[ ]	
A	**0**
B	∞
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

prev[ ]	
A	
B	
C	
D	
E	
F	
G	
H	

add $\boldsymbol{v}$ to $\boldsymbol{E}$, and set $d[v] \leftarrow \pi(v)$.

Single-source shortest paths problem

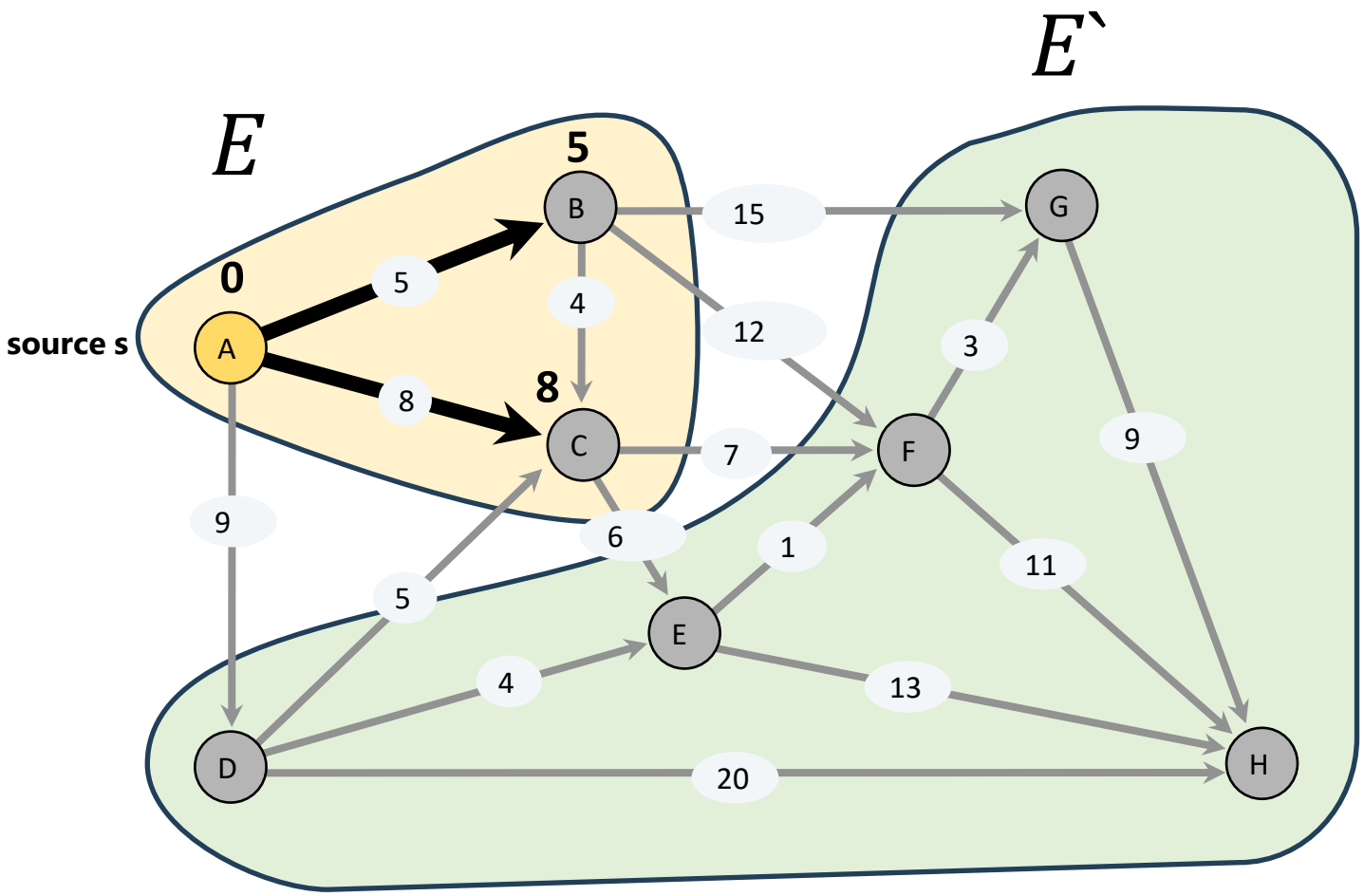


distance[ ]	
A	**0**
B	5
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

prev[ ]	
A	
B	A
C	
D	
E	
F	
G	
H	

$$\pi(v) = \min_{e=(u,v)\,:\,u \in S} d[u] + \ell_e$$

Single-source shortest paths problem

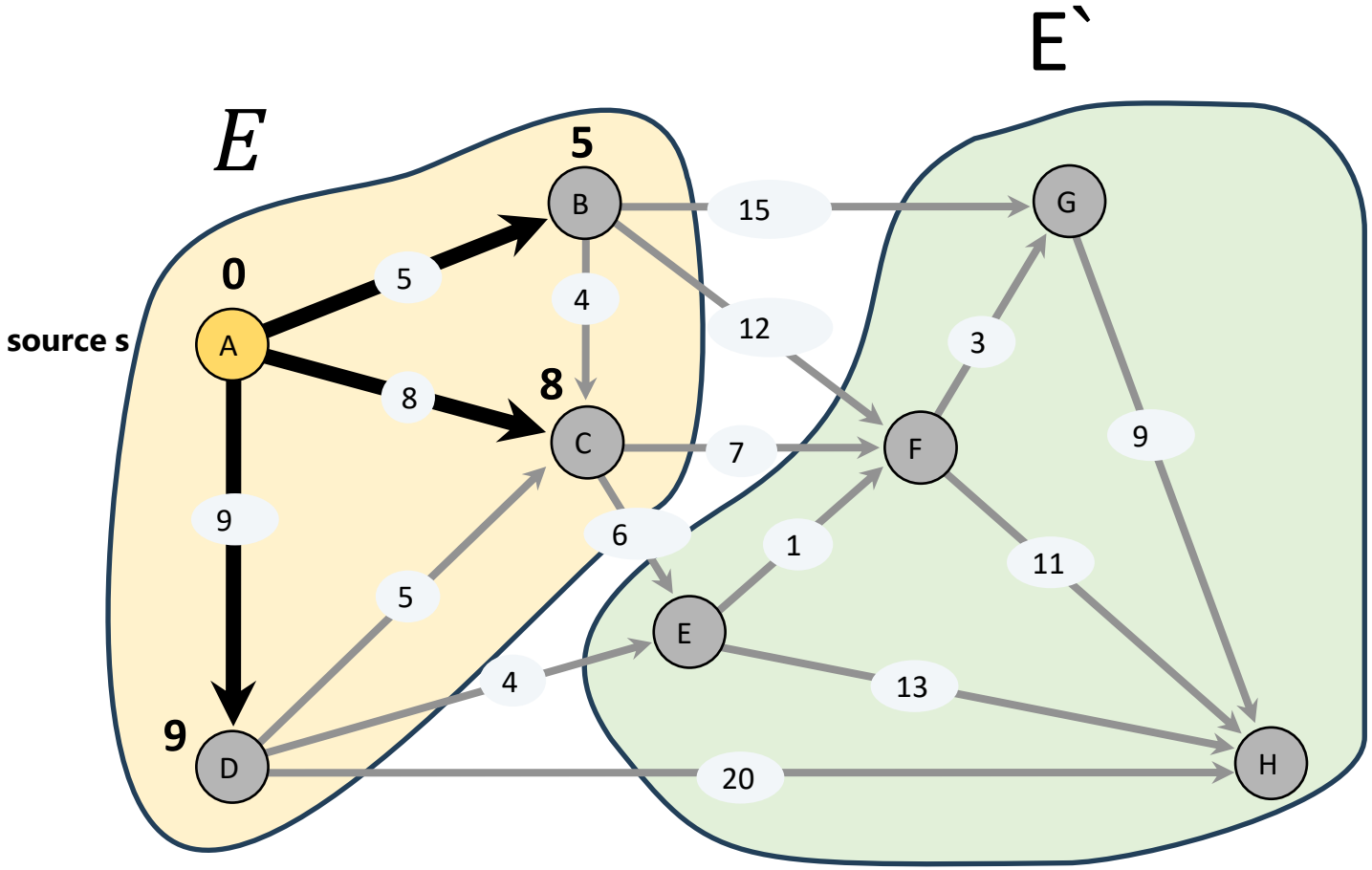


distance[ ]	
A	**0**
B	5
C	8
D	∞
E	∞
F	∞
G	∞
H	∞

prev[ ]	
A	
B	A
C	A
D	
E	
F	
G	
H	

$$\pi(v) = \min_{e=(u,v)\,:\,u \in S} d[u] + \ell_e$$

Single-source shortest paths problem

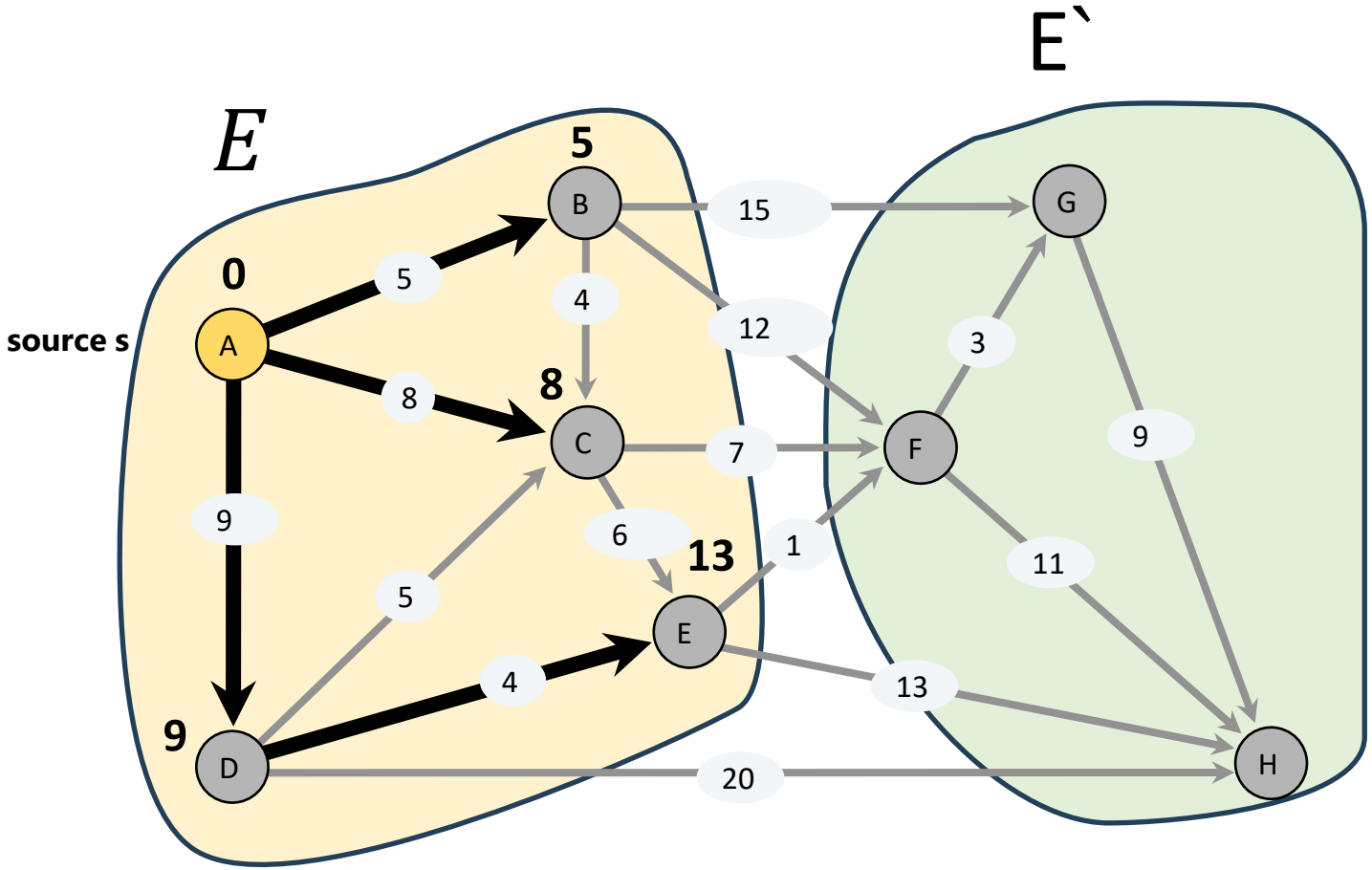


distance[ ]	
A	**0**
B	5
C	8
D	9
E	∞
F	∞
G	∞
H	∞

prev[ ]	
A	
B	A
C	A
D	A
E	
F	
G	
H	

$$\pi(v) = \min_{e=(u,v)\,:\,u\in S} d[u] + \ell_e$$

Single-source shortest paths problem



distance[ ]

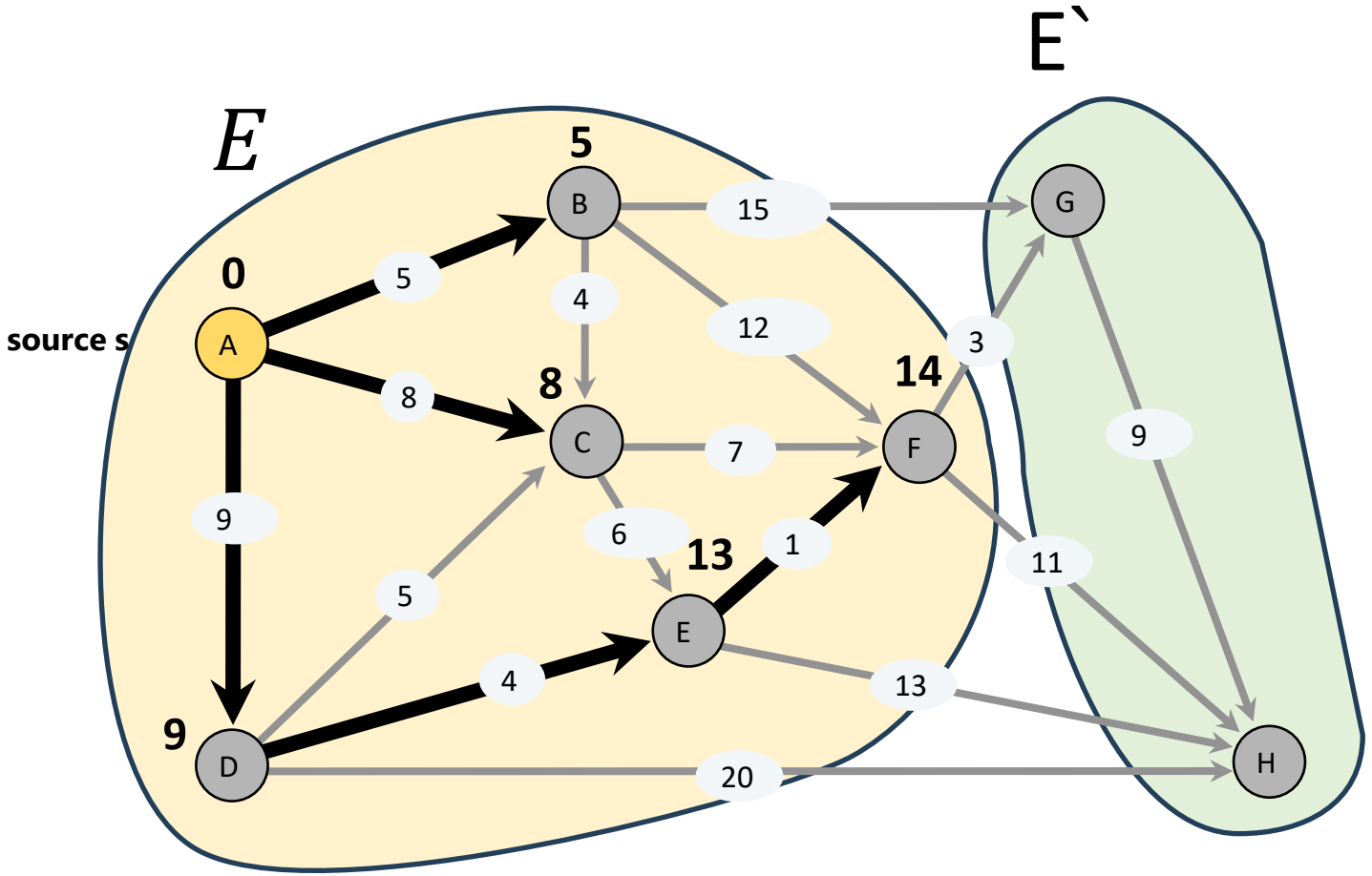
A	**0**
B	5
C	8
D	9
E	13
F	∞
G	∞
H	∞

prev[ ]

A	
B	A
C	A
D	A
E	D
F	
G	
H	

$$\pi(v) = \min_{e=(u,v)\,:\,u \in S} d[u] + \ell_e$$

Single-source shortest paths problem

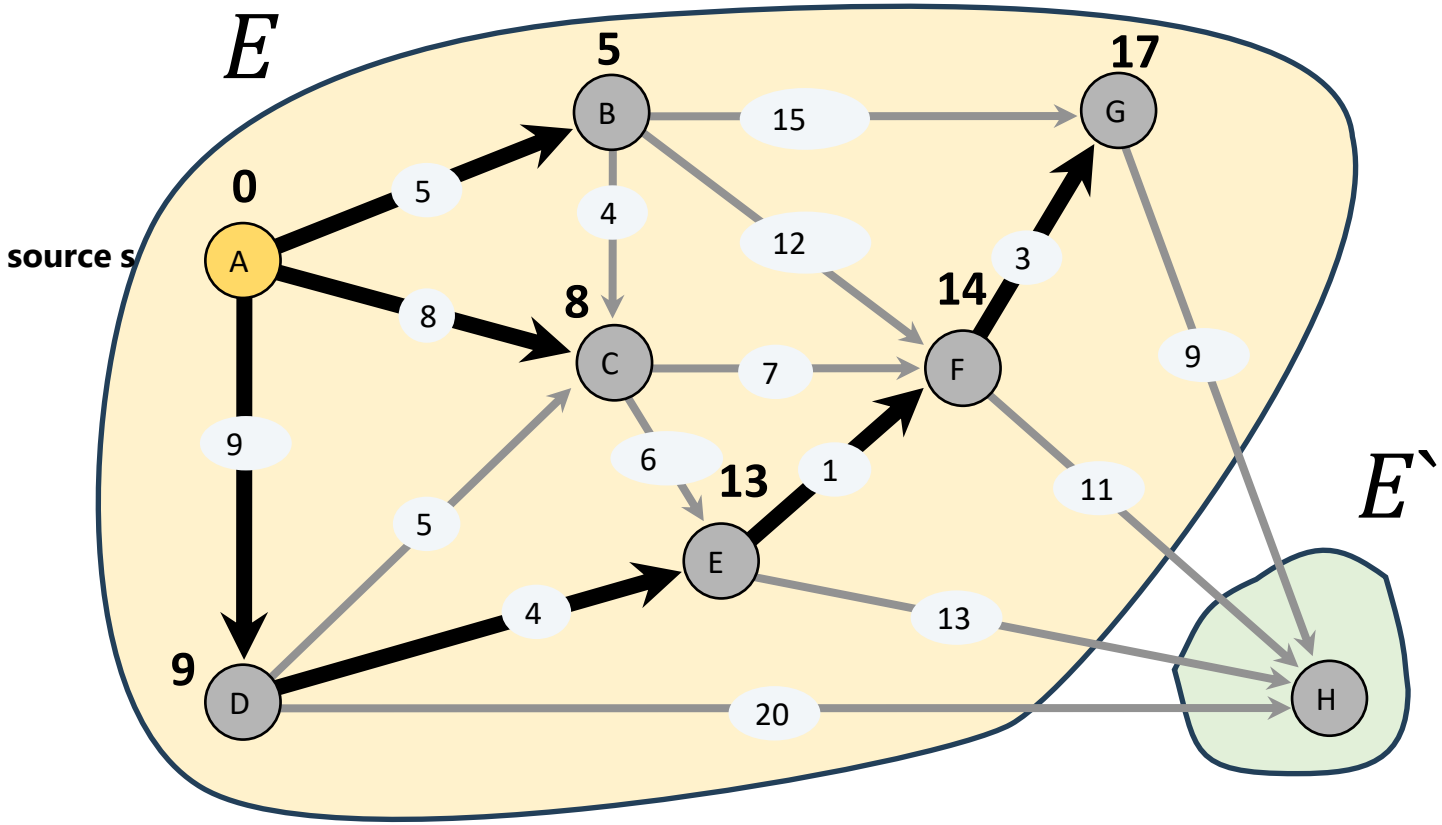


distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	∞
H	∞

prev[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	
H	

$$\pi(v) = \min_{e=(u,v)\,:\,u\in S} d[u] + \ell_e$$

Single-source shortest paths problem

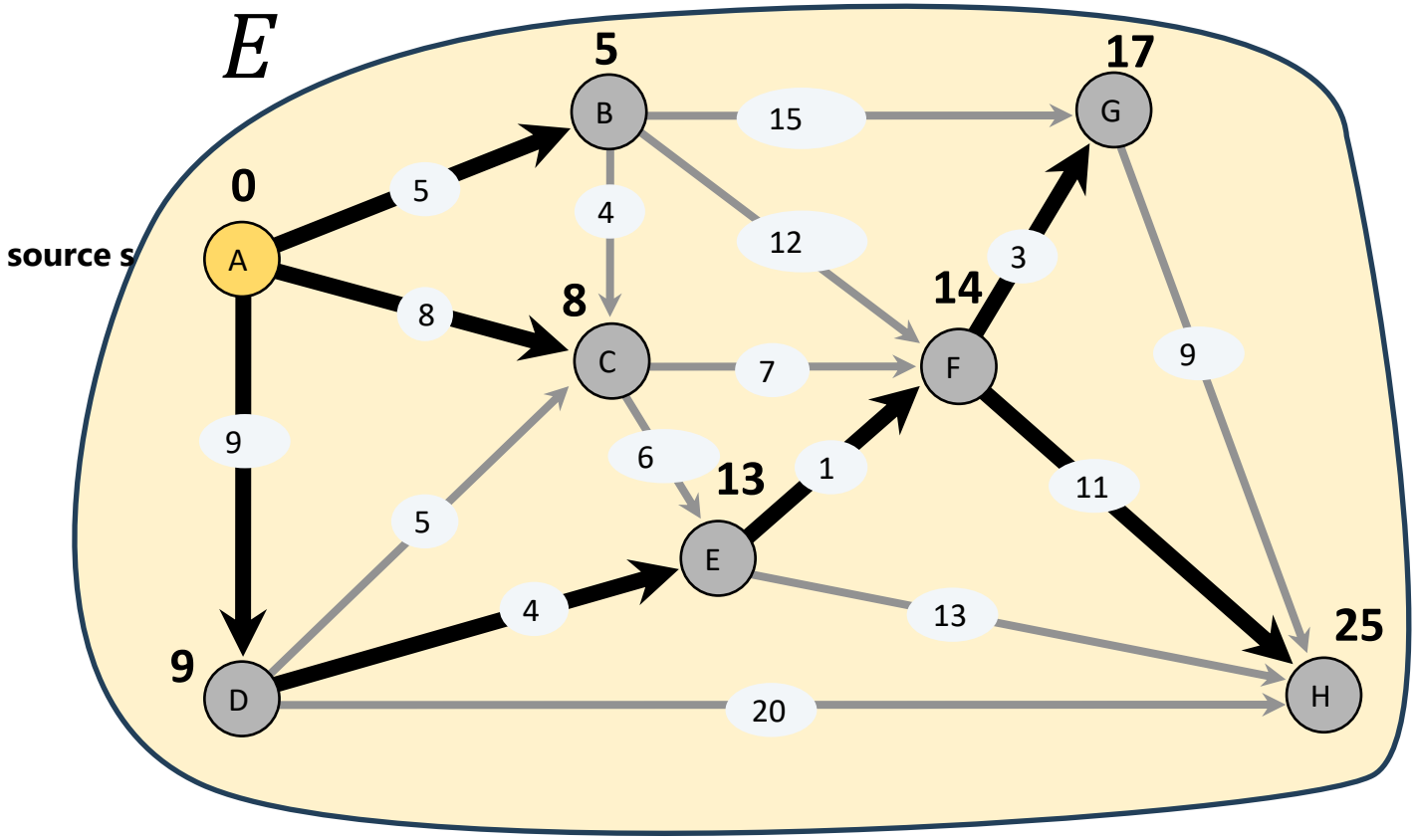


distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	17
H	∞

prev[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	F
H	

$$\pi(v) = \min_{e=(u,v)\,:\,u\in S} d[u] + \ell_e$$

Single-source shortest paths problem

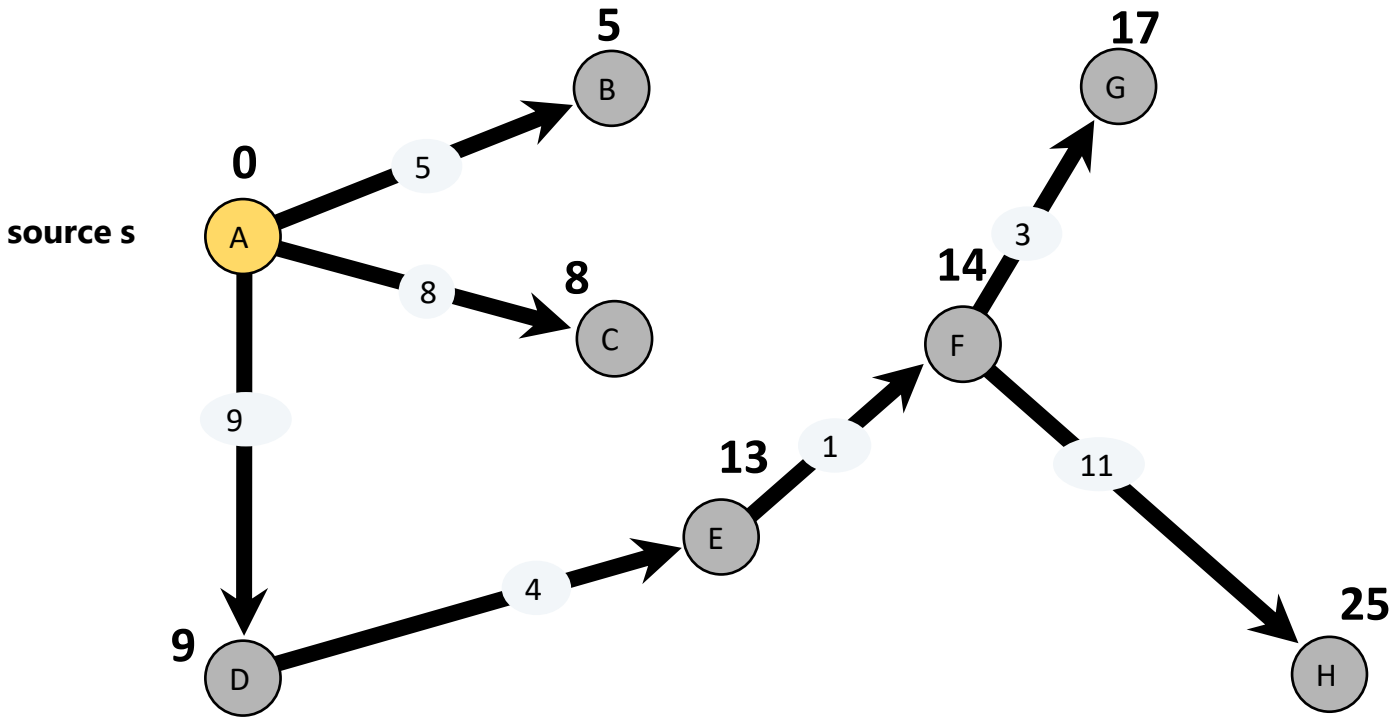


distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	17
H	25

prev[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	F
H	F

$$\pi(v) = \min_{e=(u,v)\,:\,u\in S} d[u] + \ell_e$$

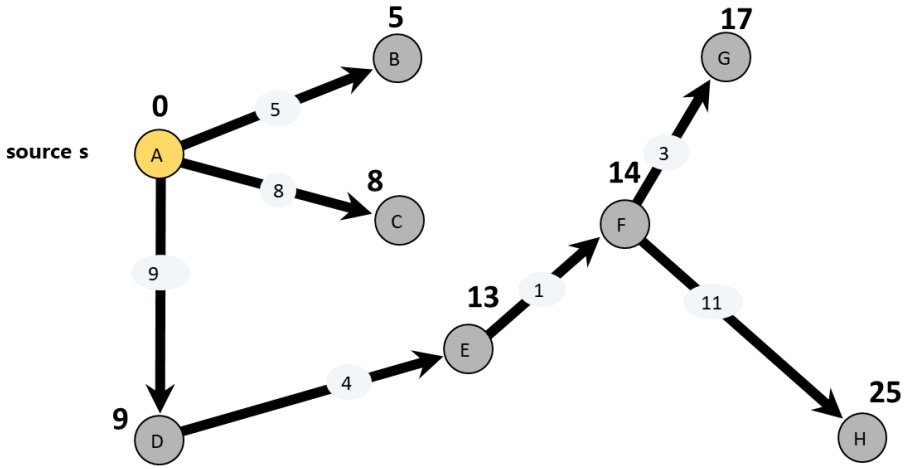
Single-source shortest paths problem



	distance[ ]
A	**0**
B	5
C	8
D	9
E	13
F	14
G	17
H	25

	prev[ ]
A	
B	A
C	A
D	A
E	D
F	E
G	F
H	F

Single-source shortest paths problem

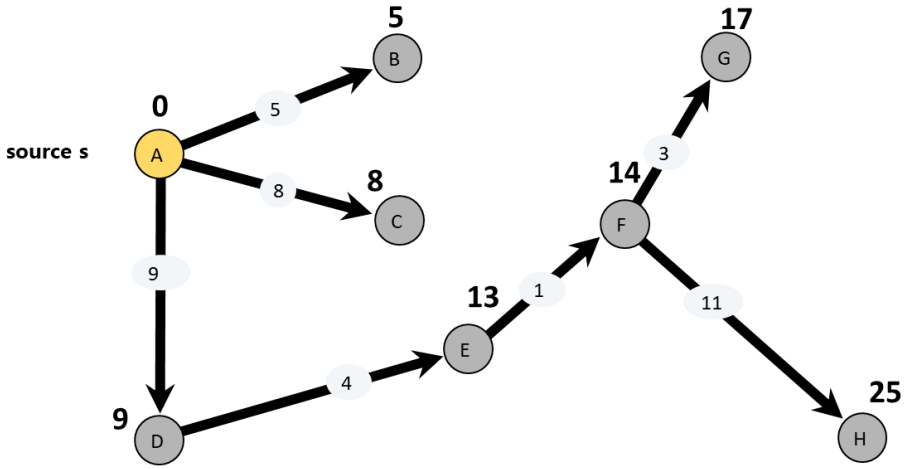


Algorithm Dijkstra's Algorithm for Shortest Paths from s to all vertices

$d[1 \ldots n] \leftarrow [\infty \ldots \infty]$
$P[1 \ldots n] \leftarrow [null \ \ldots \ null]$
$d[s] \leftarrow 0 \qquad E \leftarrow \{s\}$
while $E \neq V$ **do**
  Select $e = (u, v)$, $u \in E$, $v \notin E$ with minimum $d[u] + w(uv)$ } **O(E)**
  $E \leftarrow E \cup \{v\}$
  $d[v] \leftarrow d[u] + w(uv)$
  $P[v] \leftarrow u$

} **O(V)**

Single-source shortest paths problem



Algorithm Dijkstra's Algorithm for Shortest Paths from s to all vertices

$d[1 \ldots n] \leftarrow [\infty \ldots \infty]$
$P[1 \ldots n] \leftarrow [null \ \ldots \ null]$
$d[s] \leftarrow 0 \qquad E \leftarrow \{s\}$
while $E \neq V$ **do**
  Select $e = (u, v)$, $u \in E$, $v \notin E$ with minimum $d[u] + w(uv)$
  $E \leftarrow E \cup \{v\}$
  $d[v] \leftarrow d[u] + w(uv)$
  $P[v] \leftarrow u$

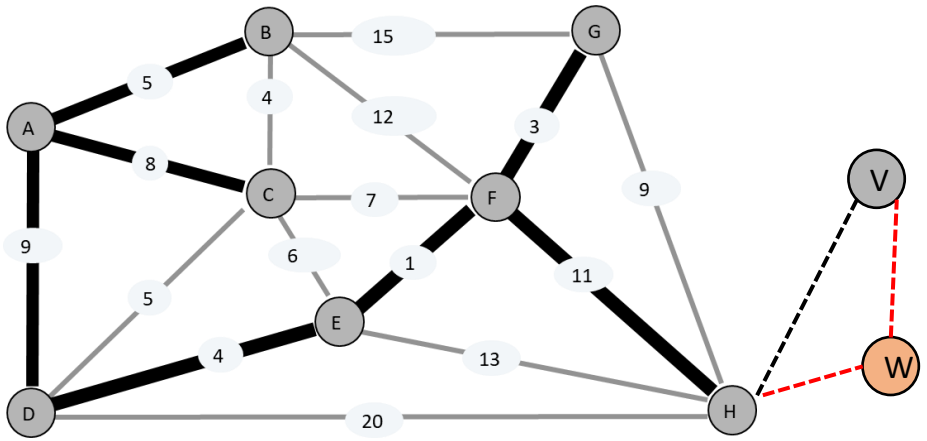
$\mathbf{O}(EV)$

Dijkstra's algorithm: Proof of correctness

- **Proof:** [by induction on $|S|$]
- **Base case:** $|S| = 1$ is easy since $S = \{ s \}$ and $d[s] = 0$.
- **Inductive hypothesis:** Assume that for some $\boldsymbol{k}$, the shortest path distance to the first $\boldsymbol{k}$ vertices added to $\boldsymbol{E}$ (let's call this subset $\boldsymbol{E_k}$) is correct. We will prove that the $\boldsymbol{d[v]}$ for the $\boldsymbol{k}$**+1th** vertex $\boldsymbol{v}$ (next closest vertex) added to $\boldsymbol{E}$ is also correct.

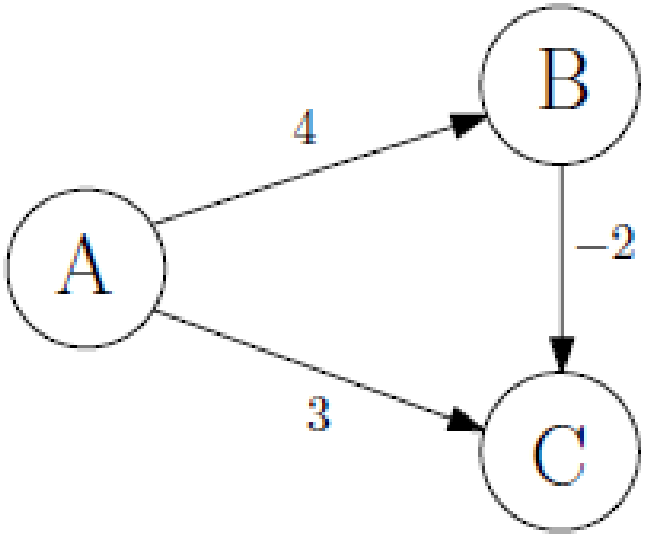
Steps:

- When vertex v added to S_k, it's chosen because it has the smallest $d[v]$ value among all vertices not in E_k
- Now, either the shortest path from s to v only goes through E_k, or it goes through at least one other node w (not in E_k)
- If the shortest path passes through w then $d[w] < d[v]$, it contradicts our inductive hypothesis
- Therefore, when the algorithm adds v to E_k, $d[v]$ correctly represents the shortest path from s to v



Live Poll 1: Dijkstra Algorithm

What paths do we get if we run Dijkstra on vertex A?



https://forms.office.com/r/bdpy1hjsK7

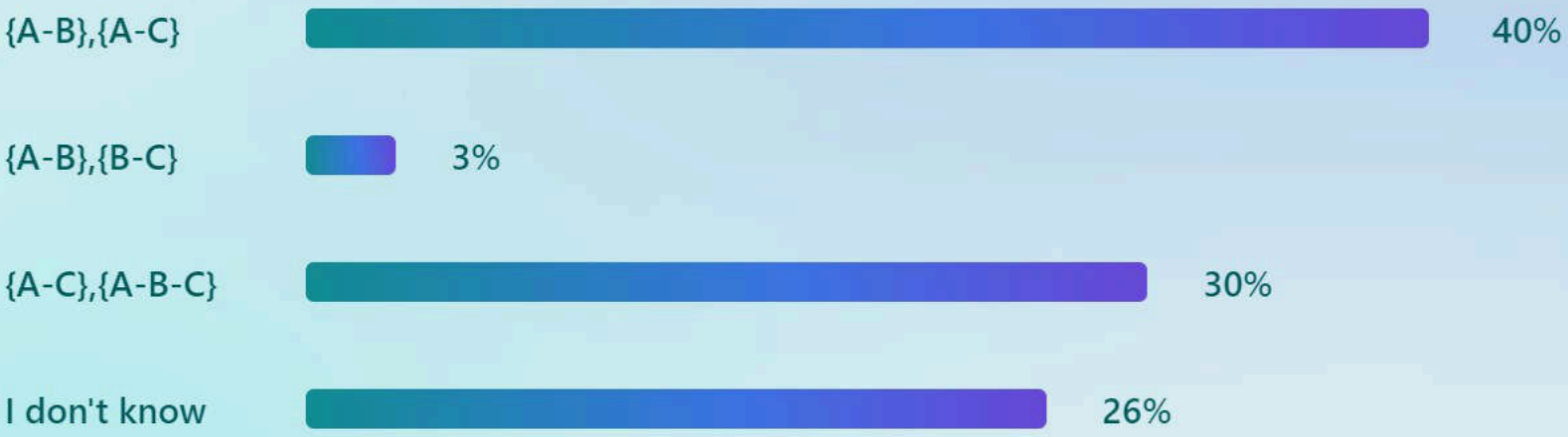
Scan the QR or use link to join

https://forms.office.com/r/bdpy1hjsK7

Copy link

31 responses submitted

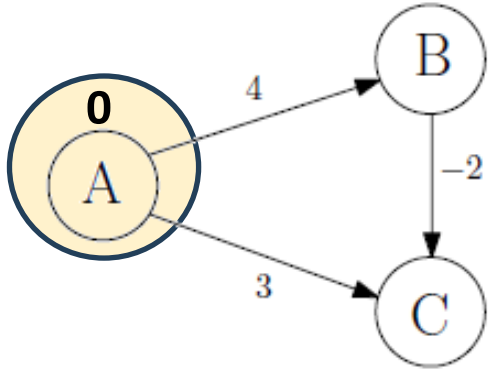
What paths do we get if we run Dijkstra on vertex A?



Treemap Bar

1 of 1

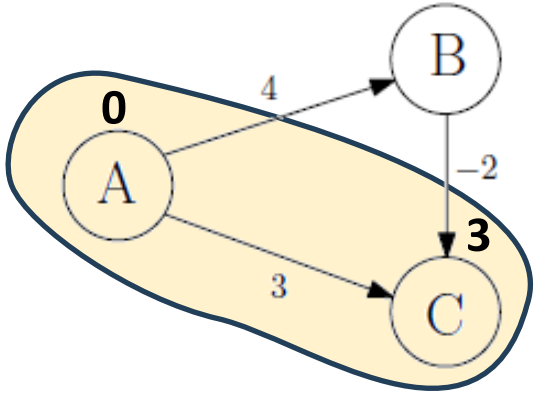
Dijkstra cannot handle *negative edges*



Algorithm Dijkstra's Algorithm for Shortest Paths from s to all vertices

$d[1 \ldots n] \leftarrow [\infty \ldots \infty]$
$P[1 \ldots n] \leftarrow [null \ \ldots \ null]$
$d[s] \leftarrow 0 \qquad E \leftarrow \{s\}$
while $E \neq V$ **do**
  Select $e = (u, v)$, $u \in E$, $v \notin E$ with minimum $d[u] + w(uv)$
  $E \leftarrow E \cup \{v\}$
  $d[v] \leftarrow d[u] + w(uv)$
  $P[v] \leftarrow u$

Dijkstra cannot handle *negative edges*



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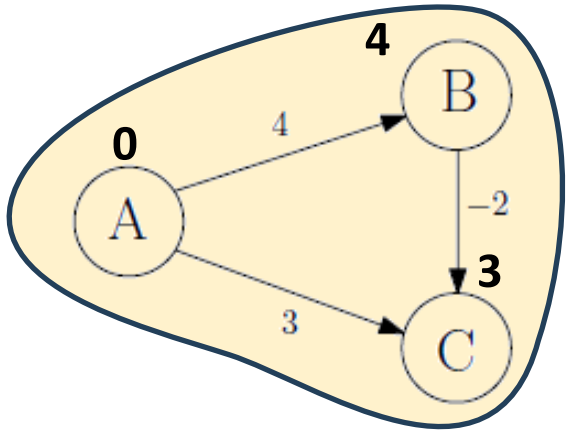
Dijkstra cannot handle *negative edges*



Algorithm Dijkstra's Algorithm for Shortest Paths from s to all vertices

$d[1 \ldots n] \leftarrow [\infty \ldots \infty]$
$P[1 \ldots n] \leftarrow [null \ \ldots \ null]$
$d[s] \leftarrow 0 \qquad E \leftarrow \{s\}$
while $E \neq V$ **do**
  Select $e = (u, v)$, $u \in E$, $v \notin E$ with minimum $d[u] + w(uv)$
  $E \leftarrow E \cup \{v\}$
  $d[v] \leftarrow d[u] + w(uv)$
  $P[v] \leftarrow u$

Can we improve the time complexity of Dijkstra?

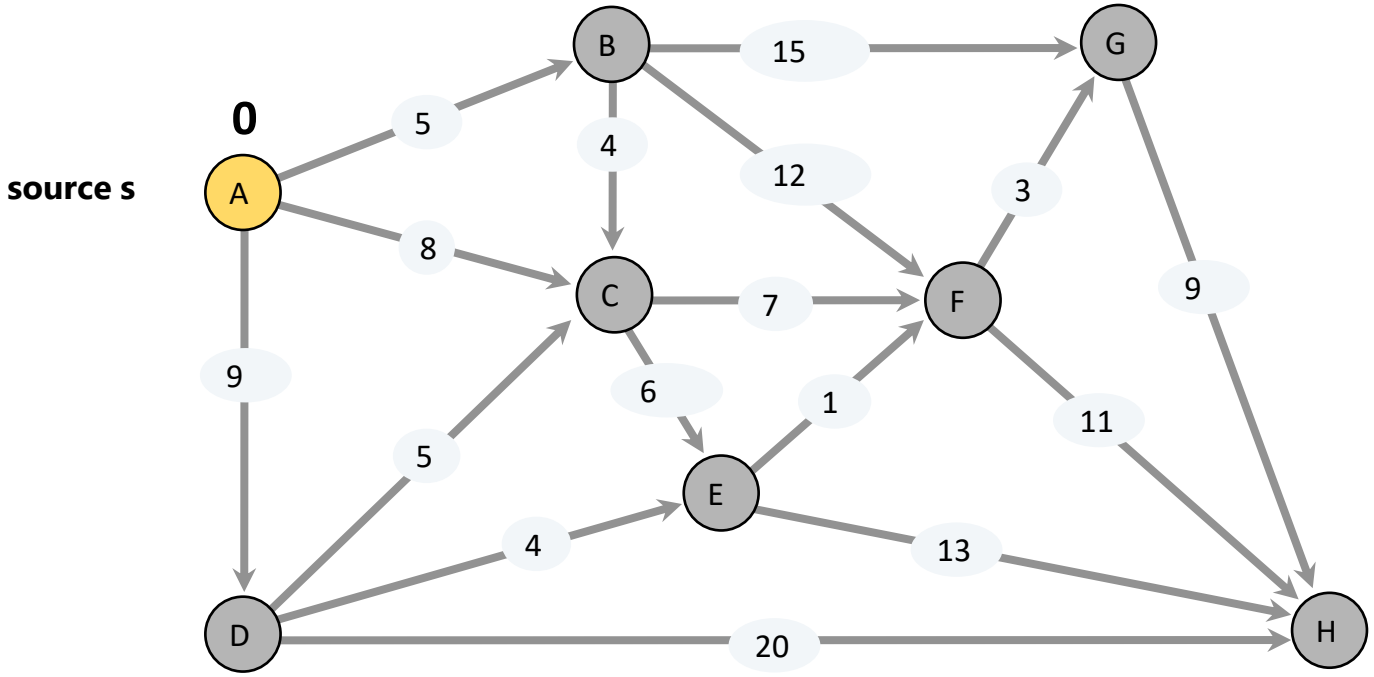
Algorithm Dijkstra's Algorithm for Shortest Paths from s to all vertices

$d[1 \ldots n] \leftarrow [\infty \ldots \infty]$
$P[1 \ldots n] \leftarrow [null \ \ldots \ null]$
$d[s] \leftarrow 0 \qquad R \leftarrow \{s\}$
while $R \neq V$ **do**
  Select $e = (u, v)$, $u \in R, v \notin R$, with minimum $d[u] + w(uv)$
  $R \leftarrow R \cup \{v\}$
  $d[v] \leftarrow d[u] + w(uv)$
  $P[v] \leftarrow u$

This operation is very expensive

Using Priority Queue

Single-source shortest paths problem



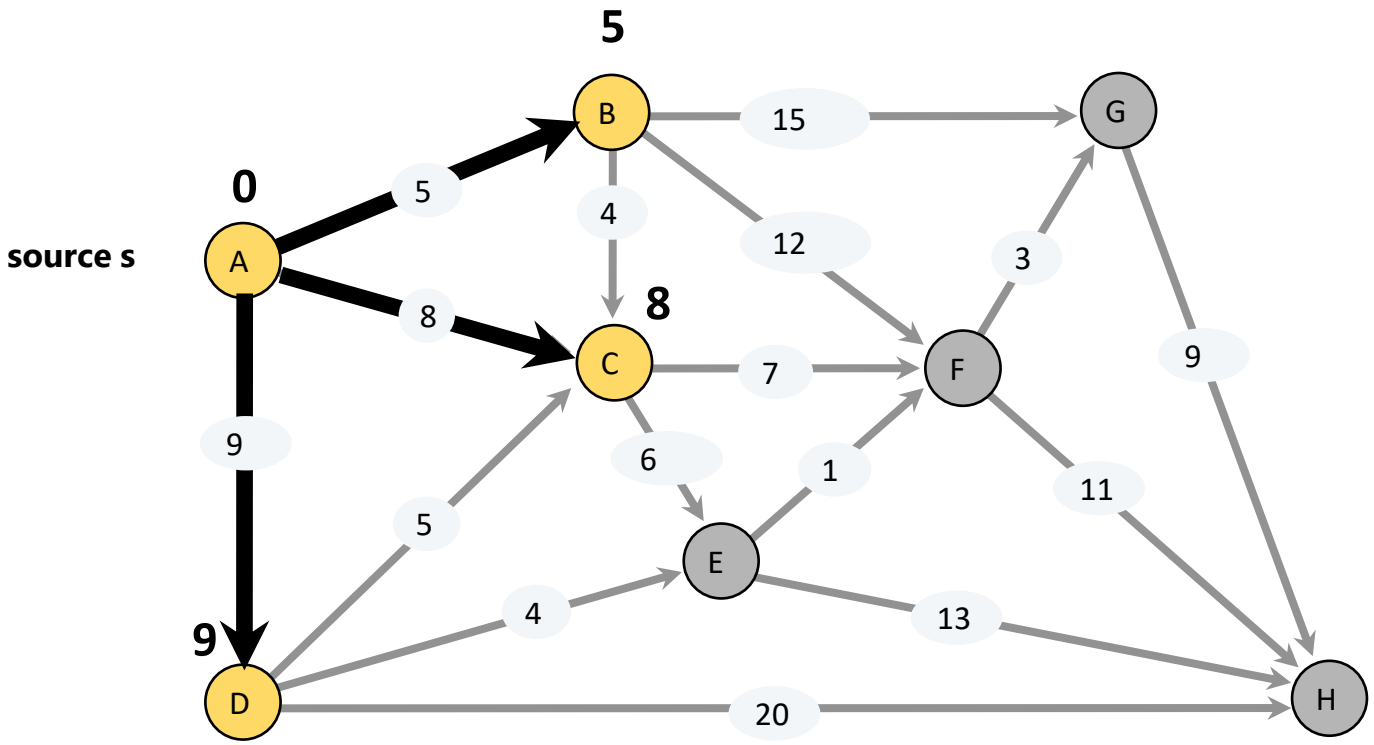
distance[ ]	
A	**0**
B	∞
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

Priority Queue	
A	**0**
B	∞
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

predecessor[ ]	
A	
B	
C	
D	
E	
F	
G	
H	

```
if (distance[u] + weight(u, v) < distance[V])
        distance[V]= distance[u] + weight(u, v)
```

Single-source shortest paths problem



distance[ ]	
A	**0**
B	5
C	8
D	9
E	∞
F	∞
G	∞
H	∞

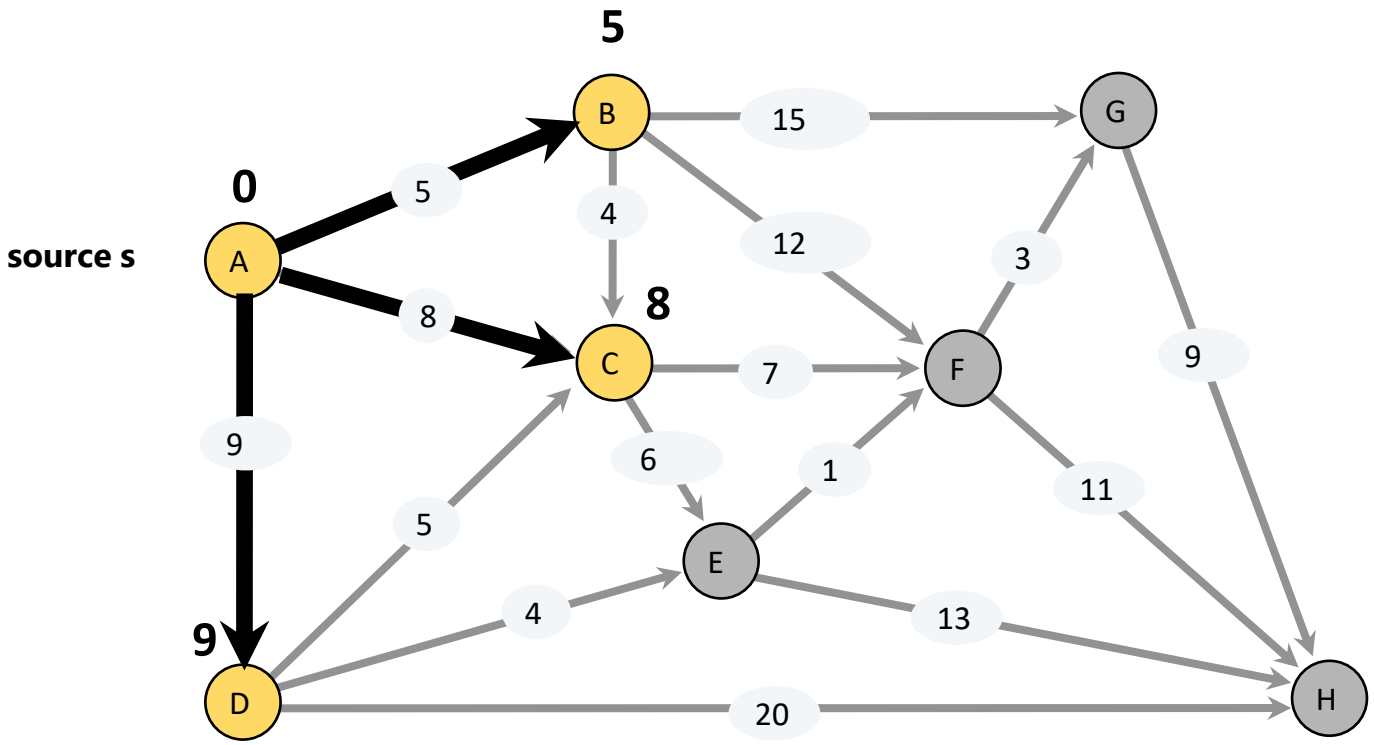
Priority Queue

B	∞
C	∞
D	∞
E	∞
F	∞
G	∞
H	∞

predecessor[ ]	
A	
B	A
C	A
D	A
E	
F	
G	
H	

```
if (distance[u] + weight(u, v) < distance[V])
        distance[V]= distance[u] + weight(u, v)
```

Single-source shortest paths problem



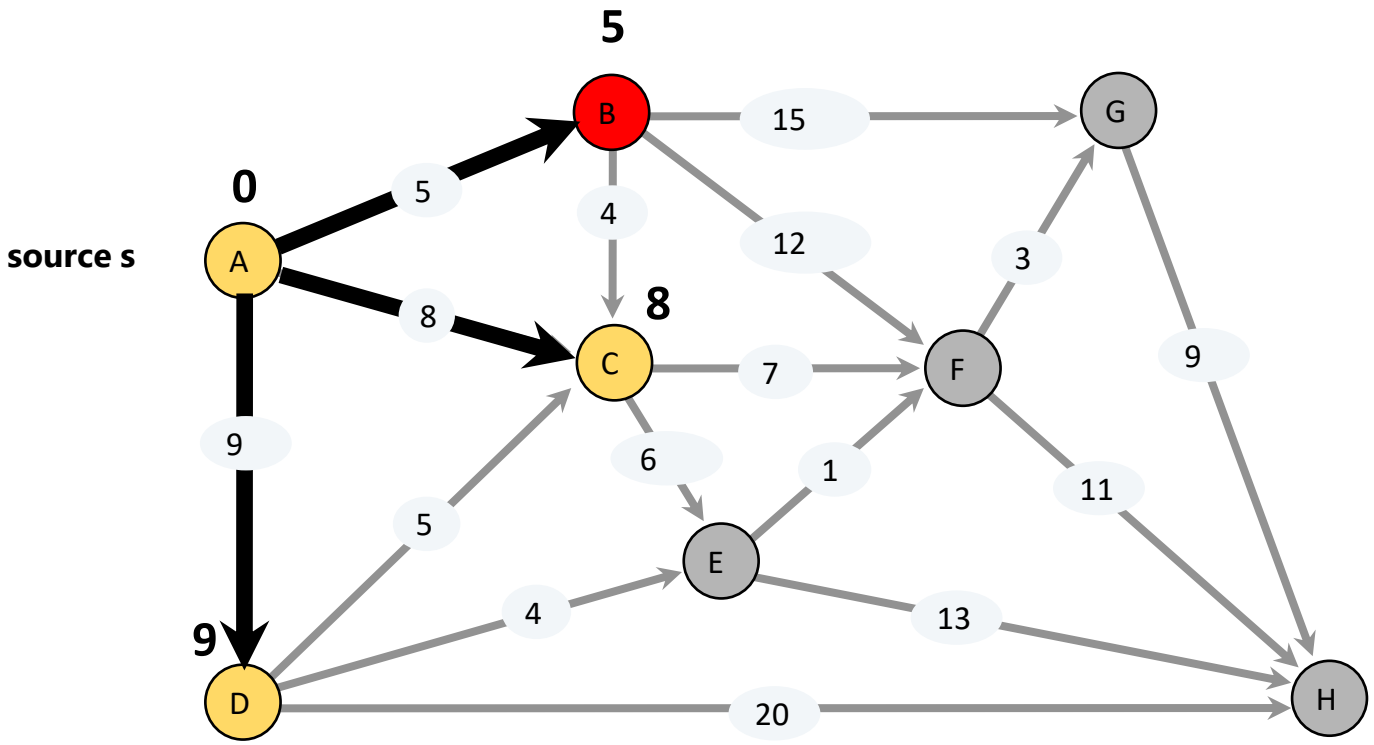
distance[ ]	
A	**0**
B	5
C	8
D	9
E	∞
F	∞
G	∞
H	∞

Priority Queue	
B	5
C	8
D	9
E	∞
F	∞
G	∞
H	∞

predecessor[ ]	
A	
B	A
C	A
D	A
E	
F	
G	
H	

```
if (distance[u] + weight(u, v) < distance[V])
        distance[V]= distance[u] + weight(u, v)
```

Single-source shortest paths problem



distance[ ]	
A	**0**
B	5
C	8
D	9
E	∞
F	∞
G	∞
H	∞

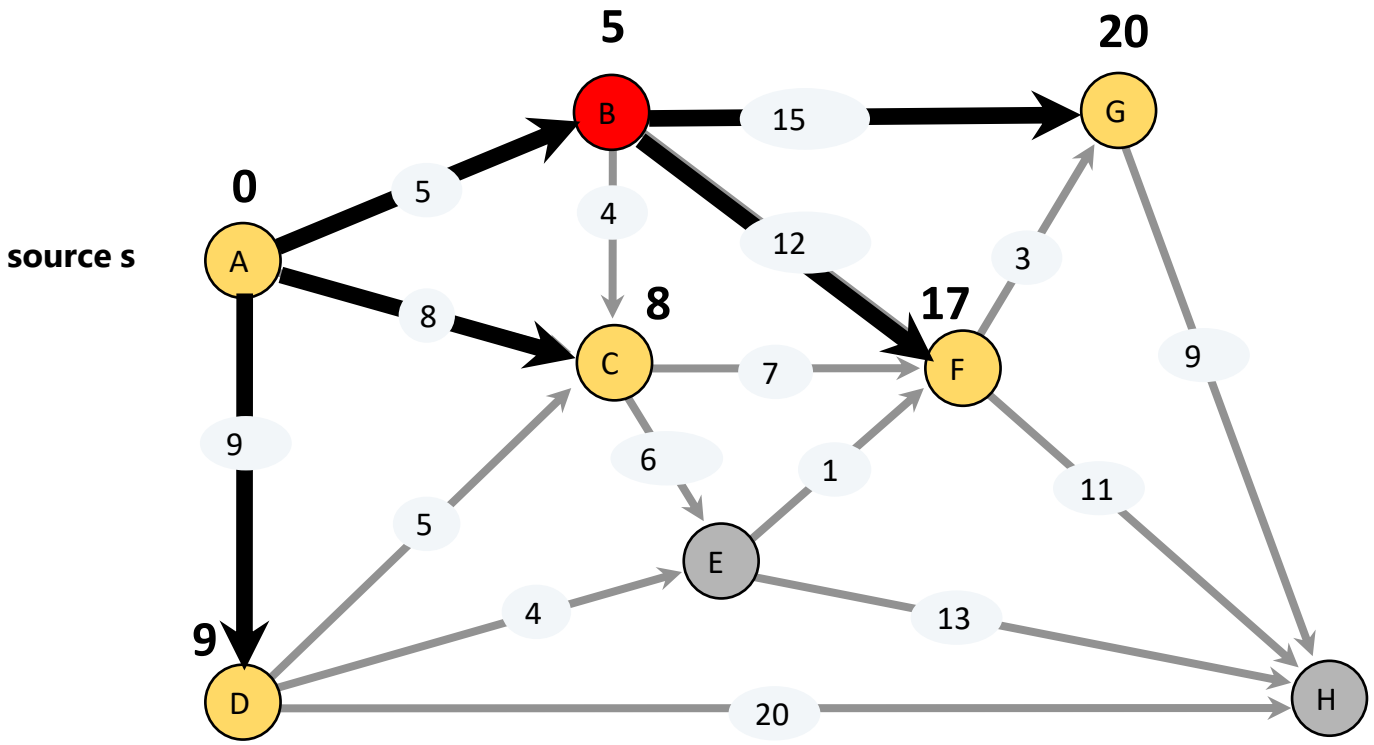
Priority Queue

C	8
D	9
E	∞
F	∞
G	∞
H	∞

predecessor[ ]	
A	
B	A
C	A
D	A
E	
F	
G	
H	

```
if (distance[u] + weight(u, v) < distance[V])
        distance[V]= distance[u] + weight(u, v)
```

Single-source shortest paths problem



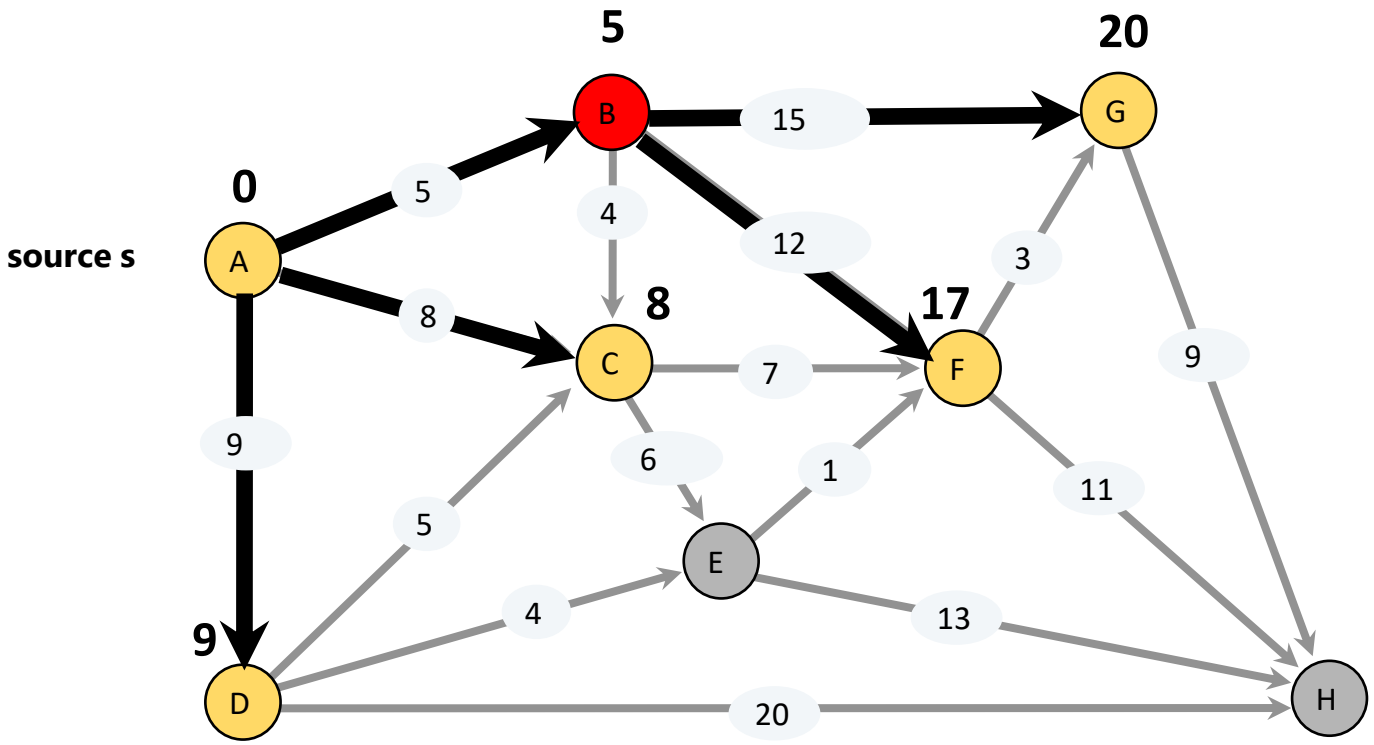
distance[ ]	
A	**0**
B	5
C	8
D	9
E	∞
F	17
G	20
H	∞

Priority Queue

C	8
D	9
E	∞
F	∞
G	∞
H	∞

predecessor[ ]	
A	
B	A
C	A
D	A
E	
F	B
G	B
H	

Single-source shortest paths problem



distance[ ]

A	**0**
B	5
C	8
D	9
E	∞
F	17
G	20
H	∞

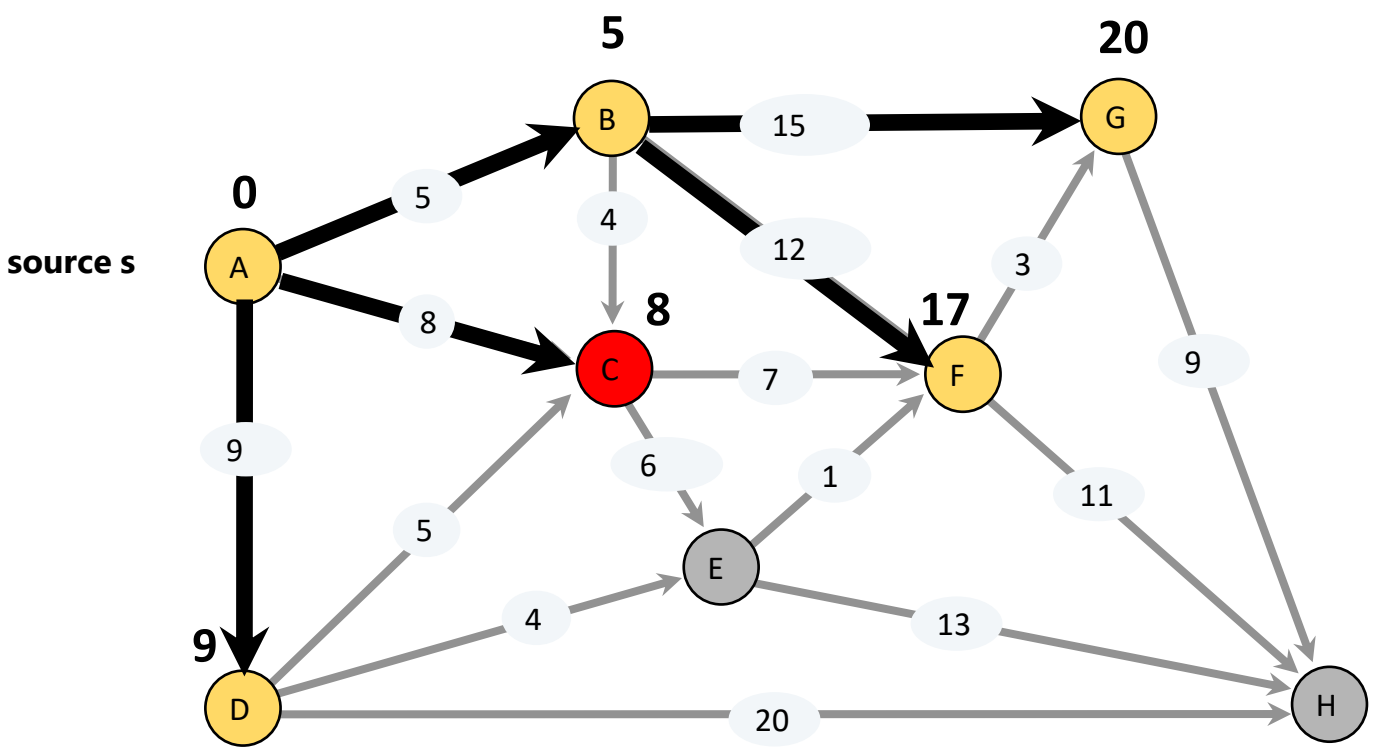
Priority Queue

C	8
D	9
F	17
G	20
E	∞
H	∞

predecessor[ ]

A	
B	A
C	A
D	A
E	
F	B
G	B
H	

Single-source shortest paths problem



distance[ ]

A	**0**
B	5
C	8
D	9
E	∞
F	17
G	20
H	∞

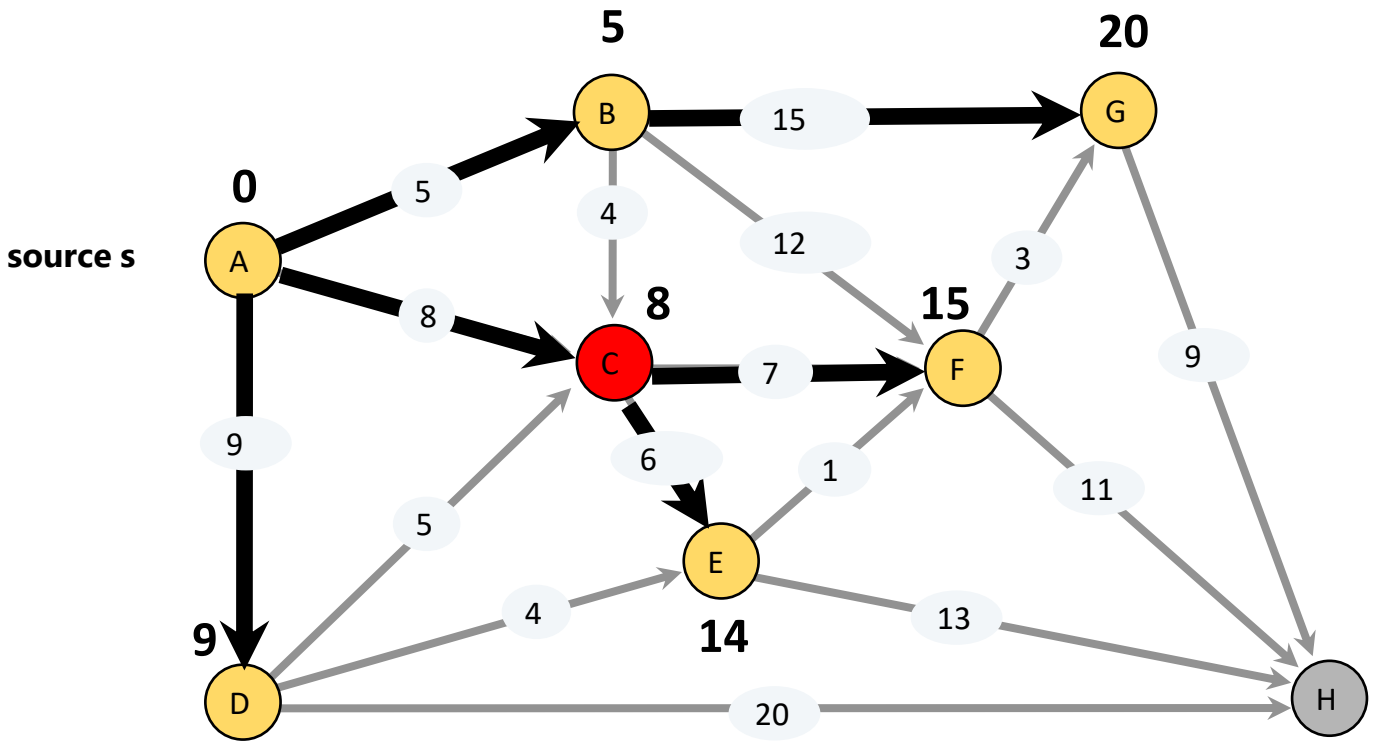
Priority Queue

D	9
F	17
G	20
E	∞
H	∞

predecessor[ ]

A	
B	A
C	A
D	A
E	
F	B
G	B
H	

Single-source shortest paths problem



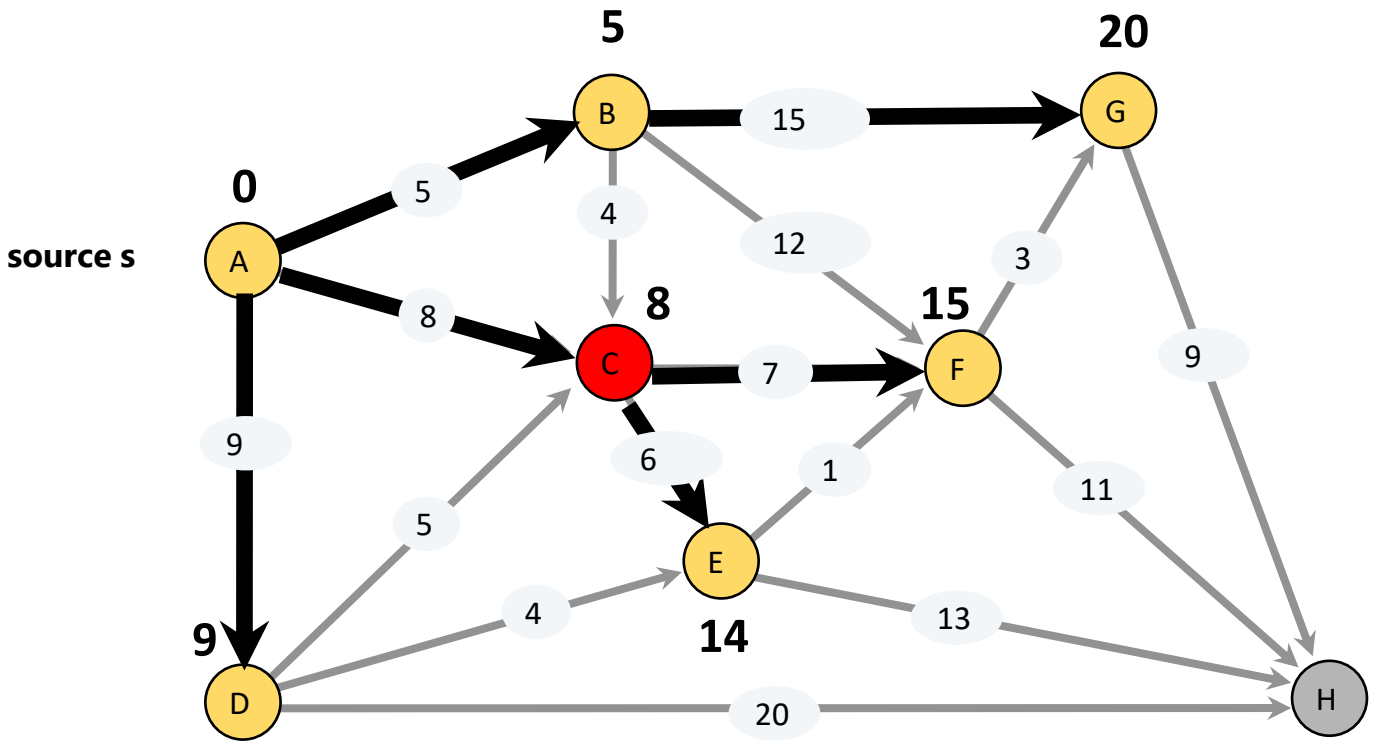
distance[ ]	
A	**0**
B	5
C	8
D	9
E	14
F	15
G	20
H	∞

Priority Queue

D	9
F	17
G	20
E	∞
H	∞

predecessor[ ]	
A	
B	A
C	A
D	A
E	C
F	C
G	B
H	

Single-source shortest paths problem



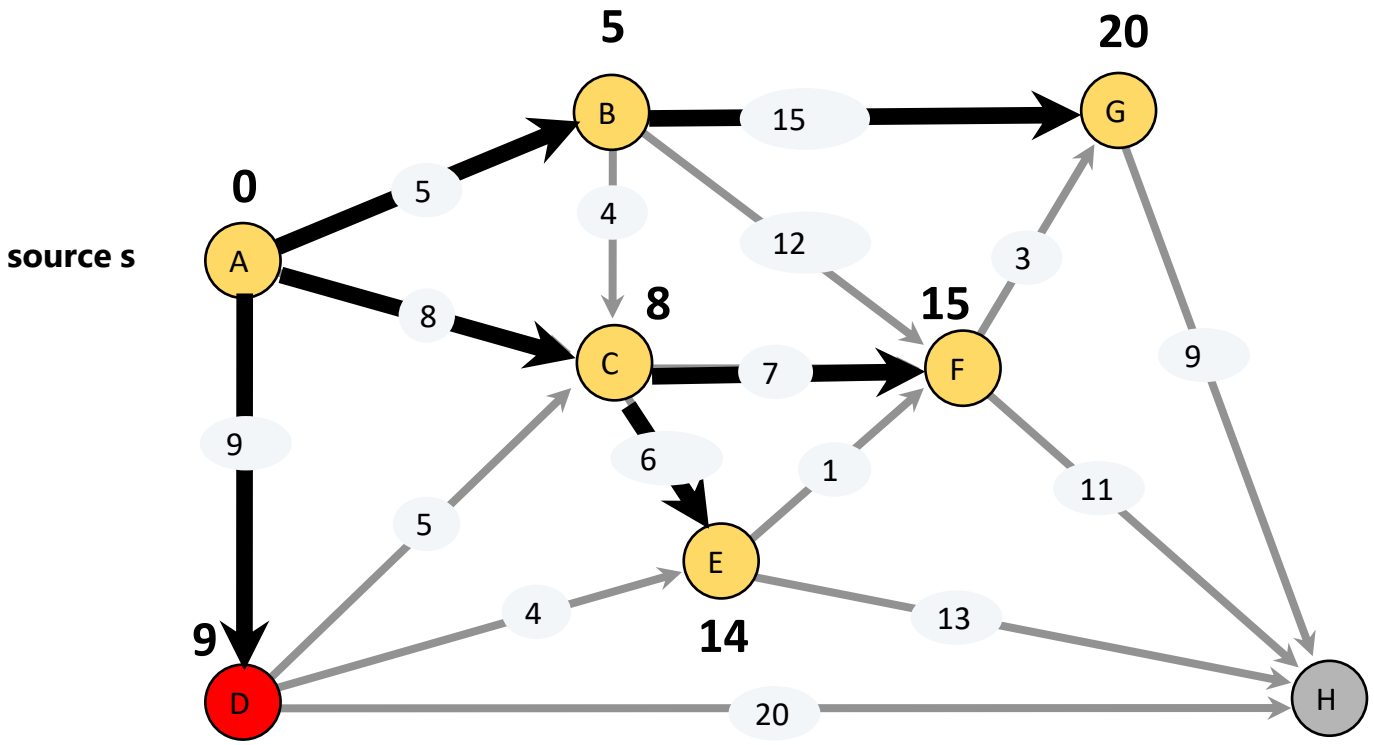
distance[ ]	
A	**0**
B	5
C	8
D	9
E	14
F	15
G	20
H	∞

Priority Queue

D	9
E	14
F	15
G	20
H	∞

predecessor[ ]	
A	
B	A
C	A
D	A
E	C
F	C
G	B
H	

Single-source shortest paths problem



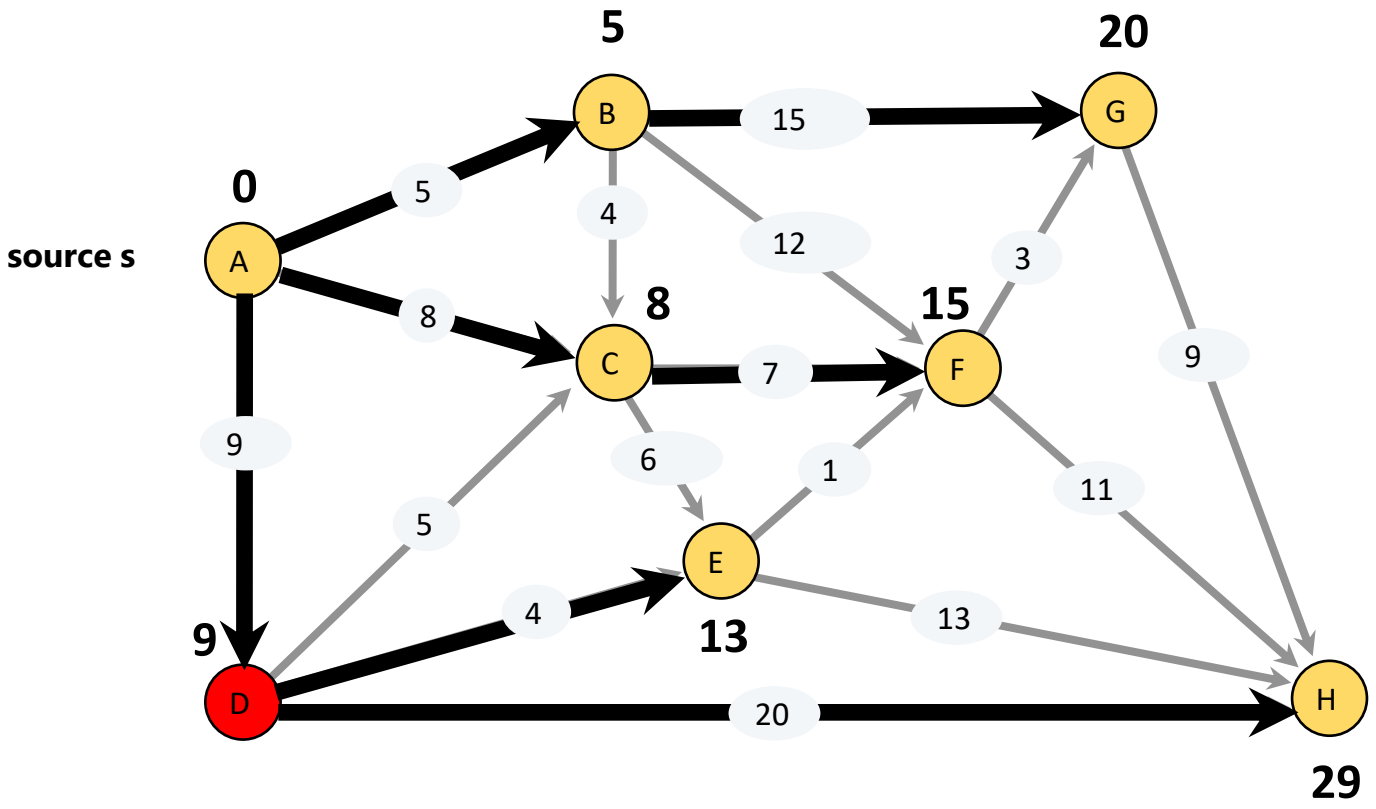
distance[ ]	
A	**0**
B	5
C	8
D	9
E	14
F	15
G	20
H	∞

Priority Queue

E	14
F	15
G	20
H	∞

predecessor[ ]	
A	
B	A
C	A
D	A
E	C
F	C
G	B
H	

Single-source shortest paths problem



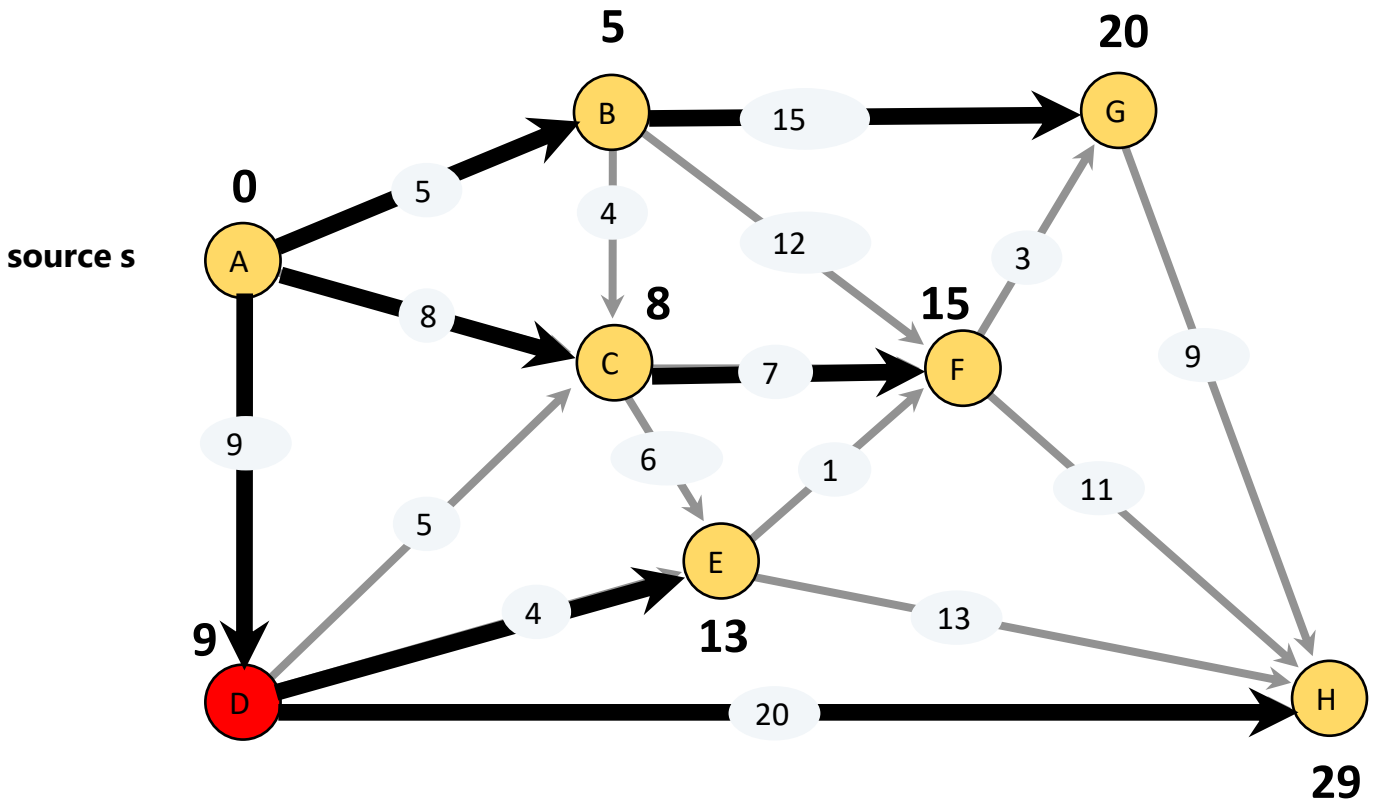
distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	15
G	20
H	29

Priority Queue

E	14
F	15
G	20
H	∞

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	C
G	B
H	D

Single-source shortest paths problem



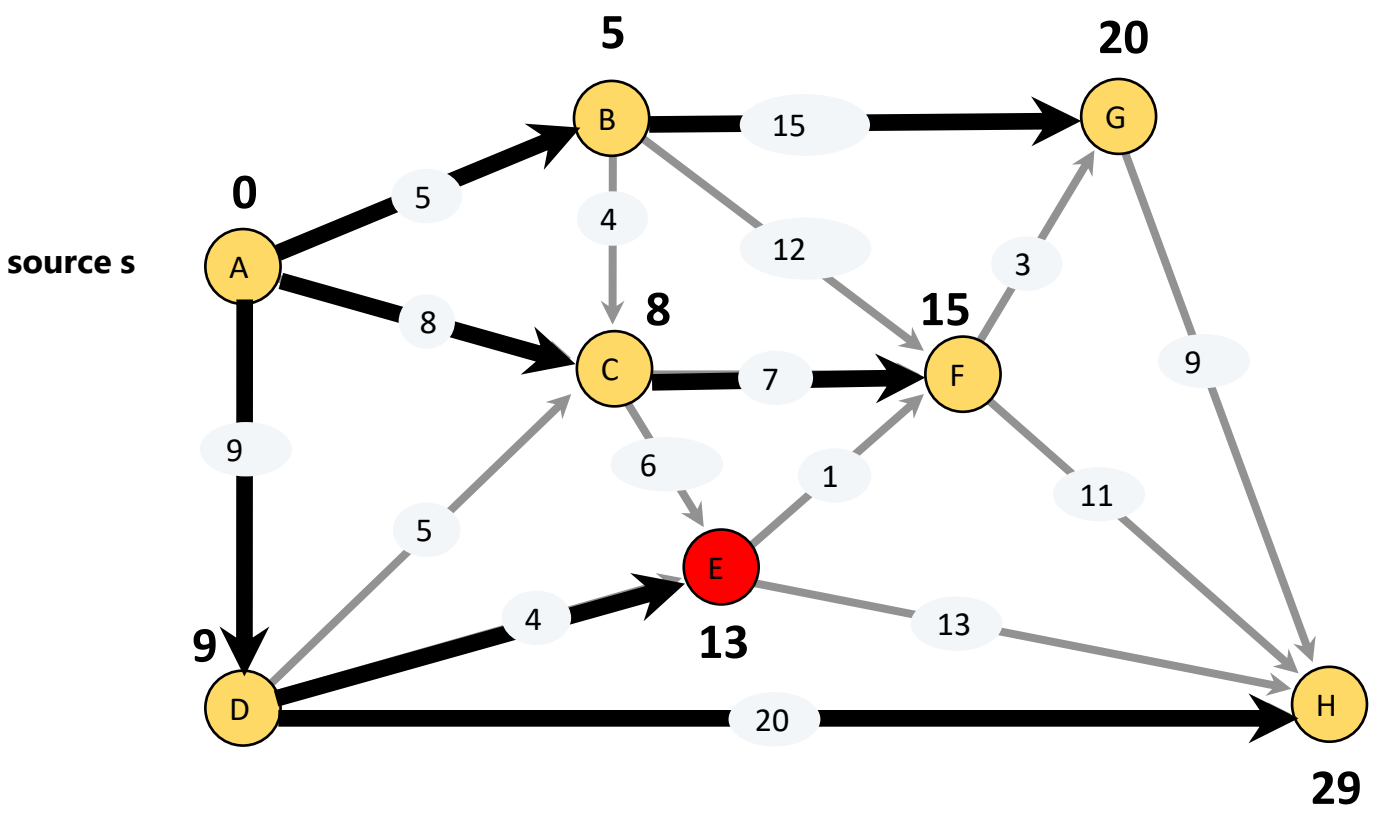
distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	15
G	20
H	29

Priority Queue

E	13
F	15
G	20
H	29

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	C
G	B
H	D

Single-source shortest paths problem



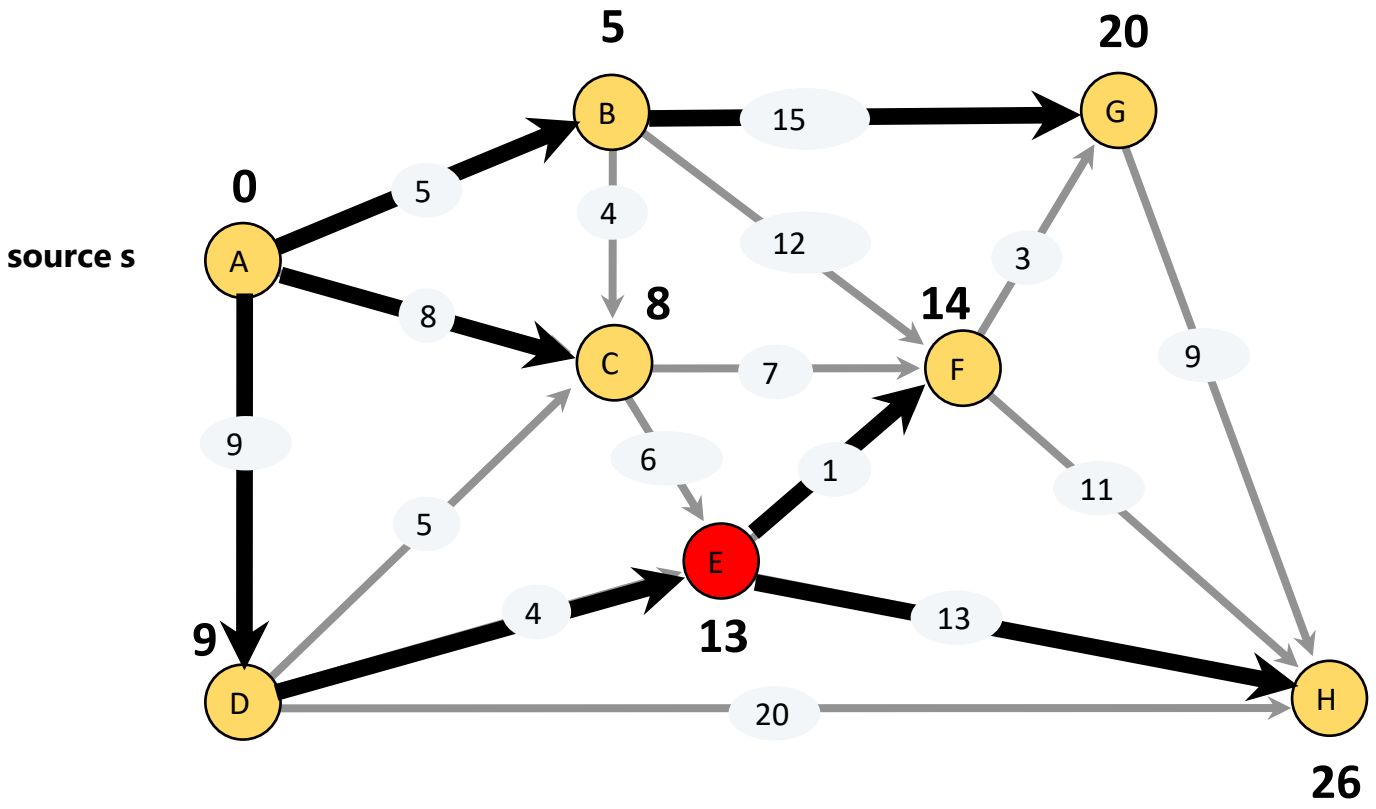
distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	15
G	20
H	29

Priority Queue

F	15
G	20
H	29

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	C
G	B
H	D

Single-source shortest paths problem



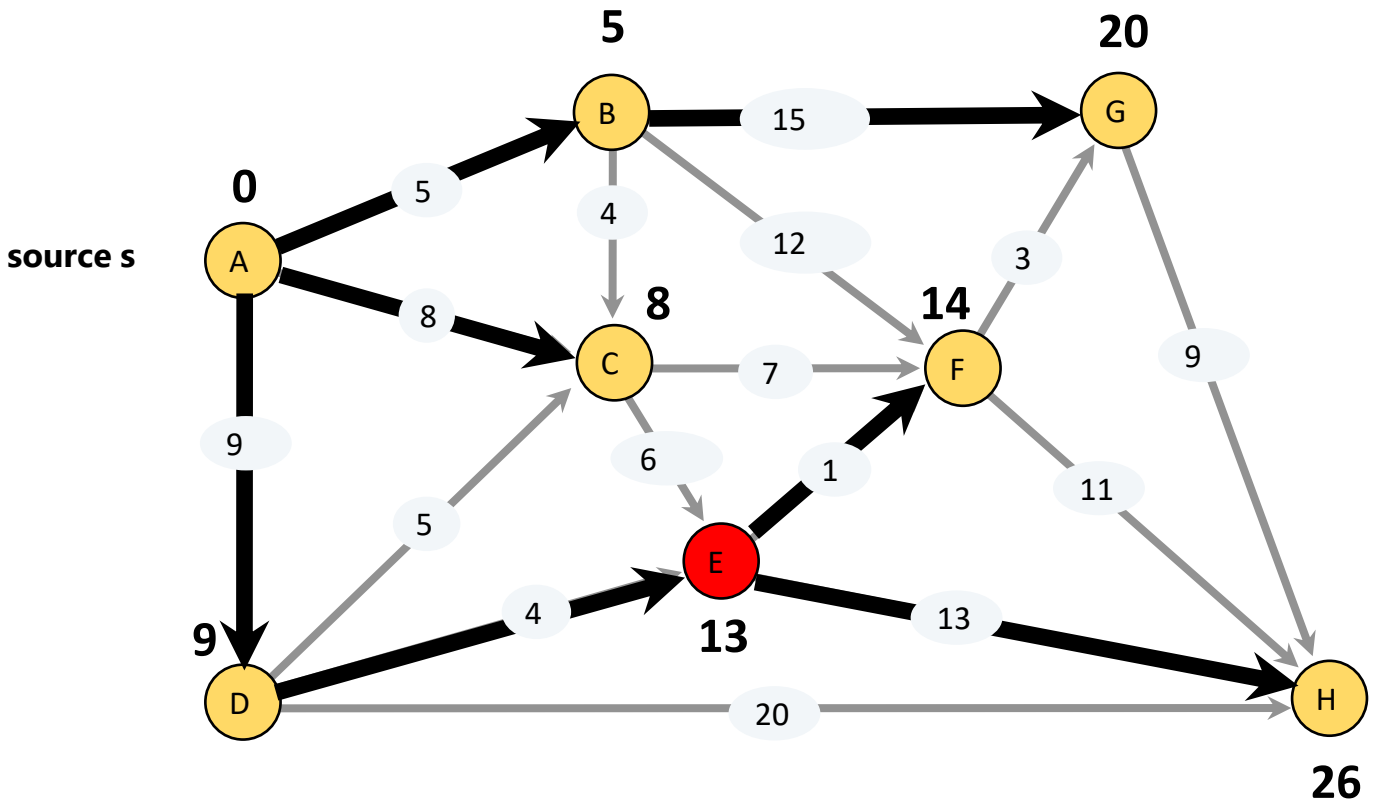
distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	20
H	26

Priority Queue

F	15
G	20
H	29

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	B
H	E

Single-source shortest paths problem



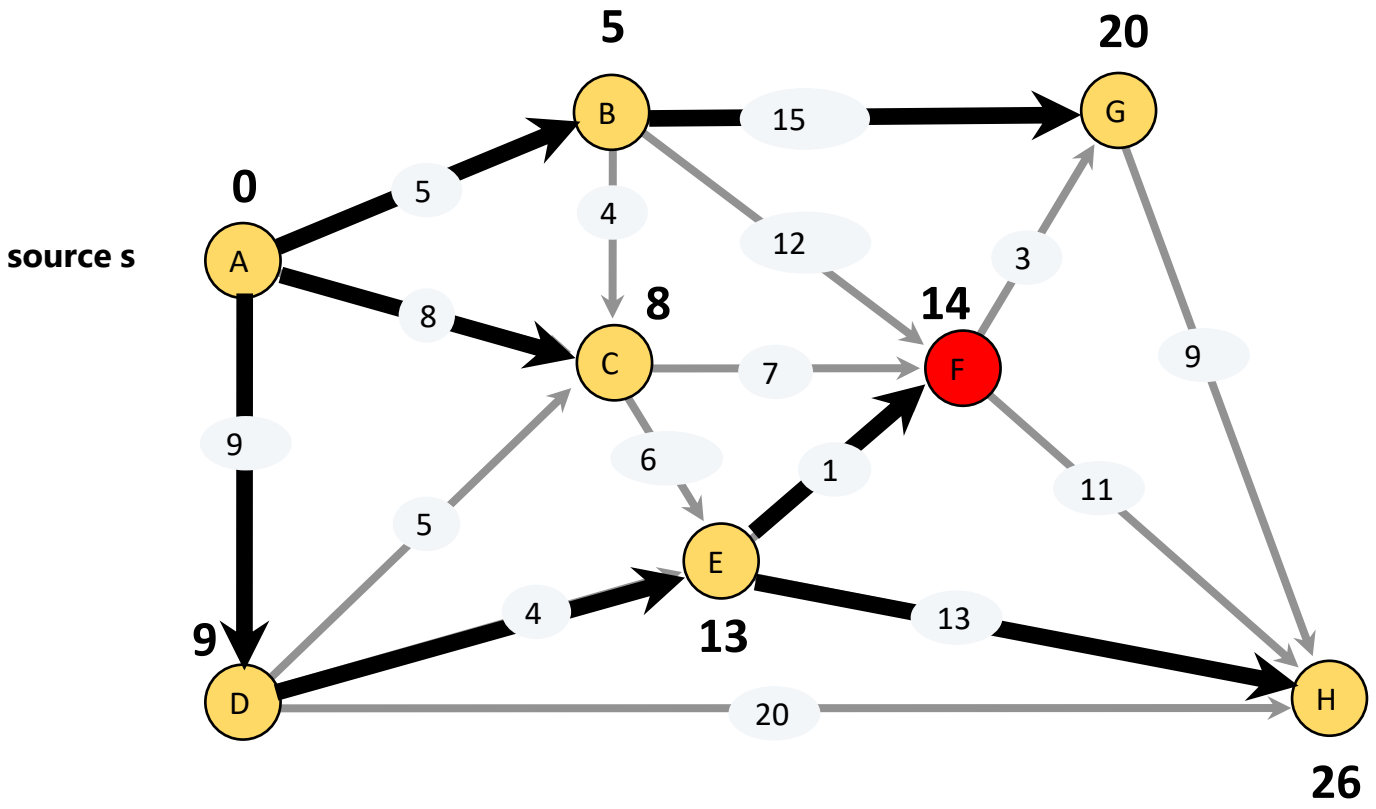
distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	20
H	26

Priority Queue

F	14
G	20
H	26

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	B
H	E

Single-source shortest paths problem



distance[ ]

A	**0**
B	5
C	8
D	9
E	13
F	14
G	20
H	26

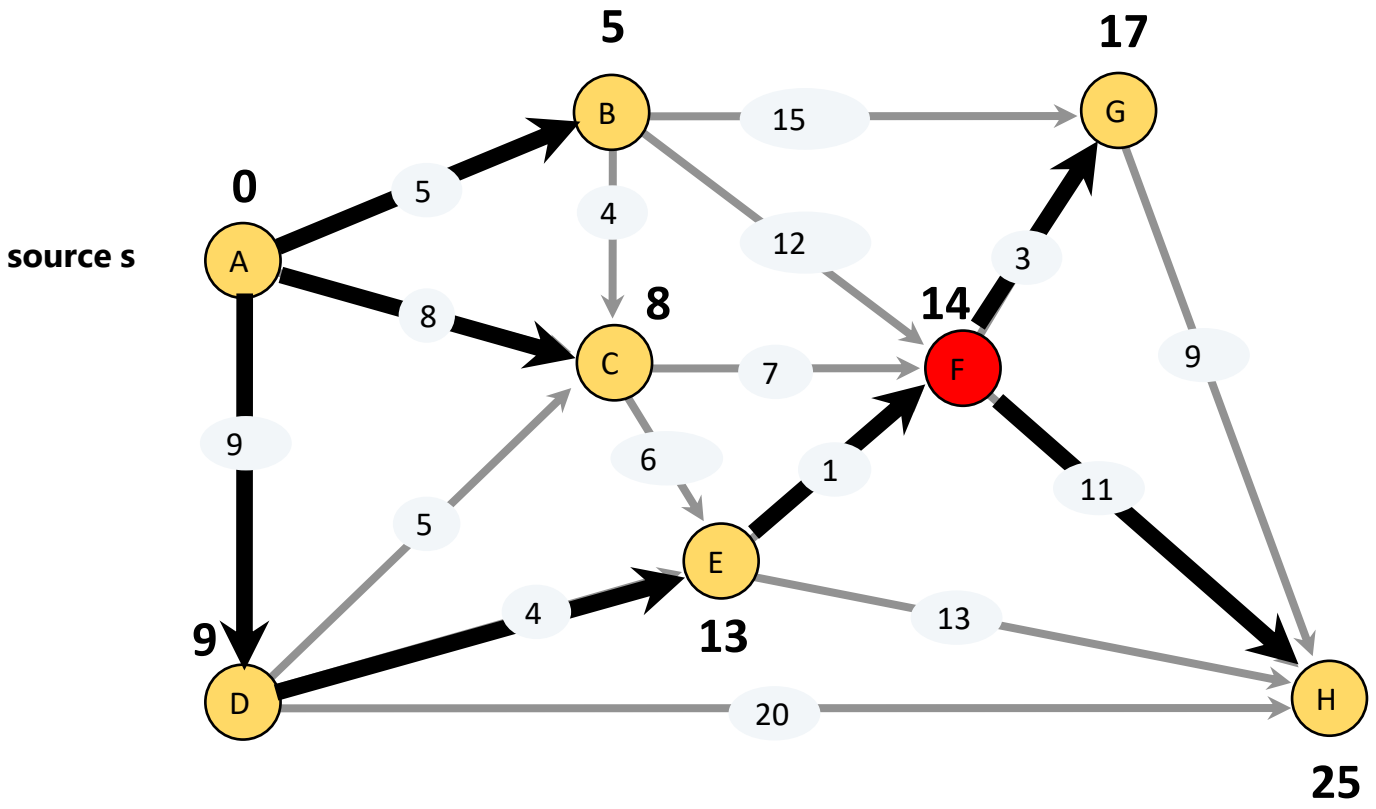
Priority Queue

G	20
H	26

predecessor[ ]

A	
B	A
C	A
D	A
E	D
F	E
G	B
H	E

Single-source shortest paths problem



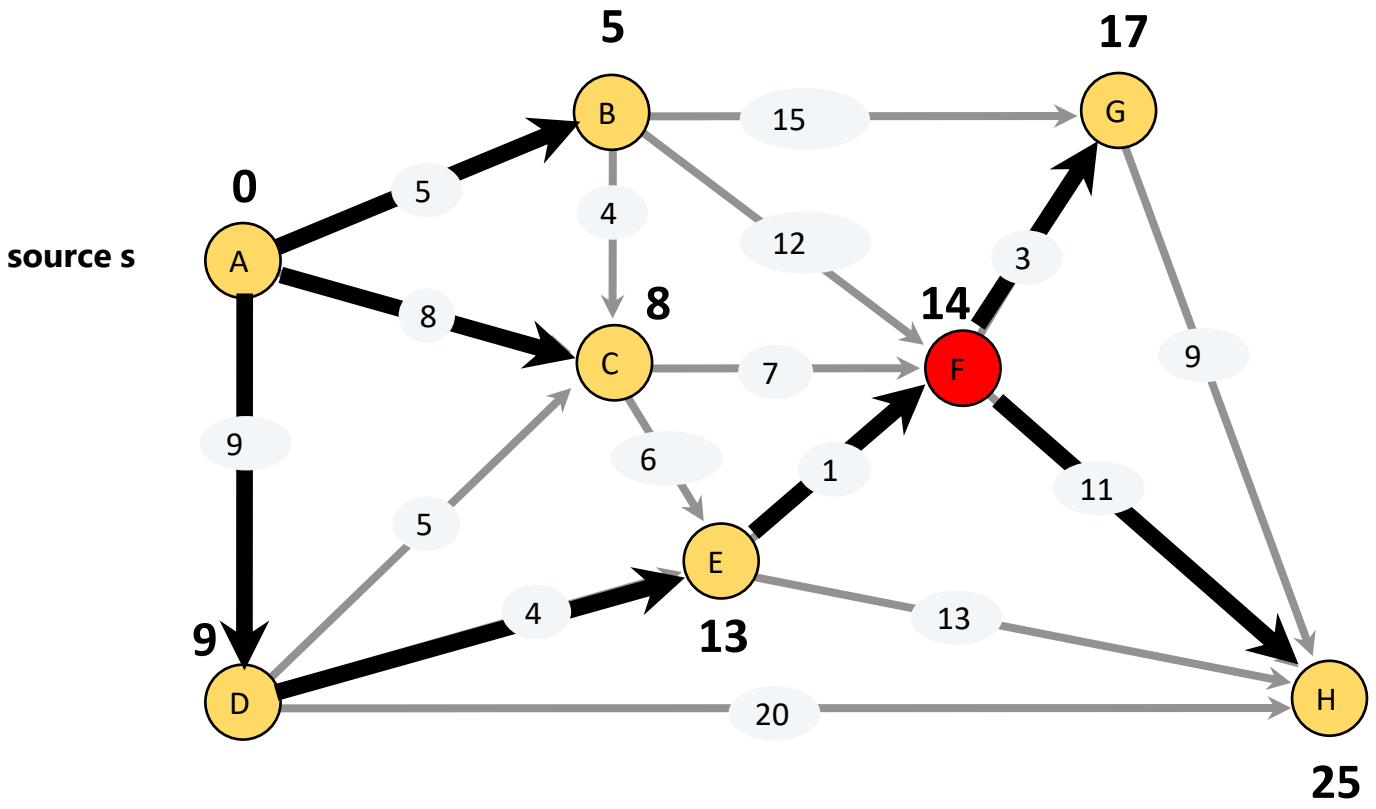
distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	17
H	25

Priority Queue

G	20
H	26

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	F
H	F

Single-source shortest paths problem



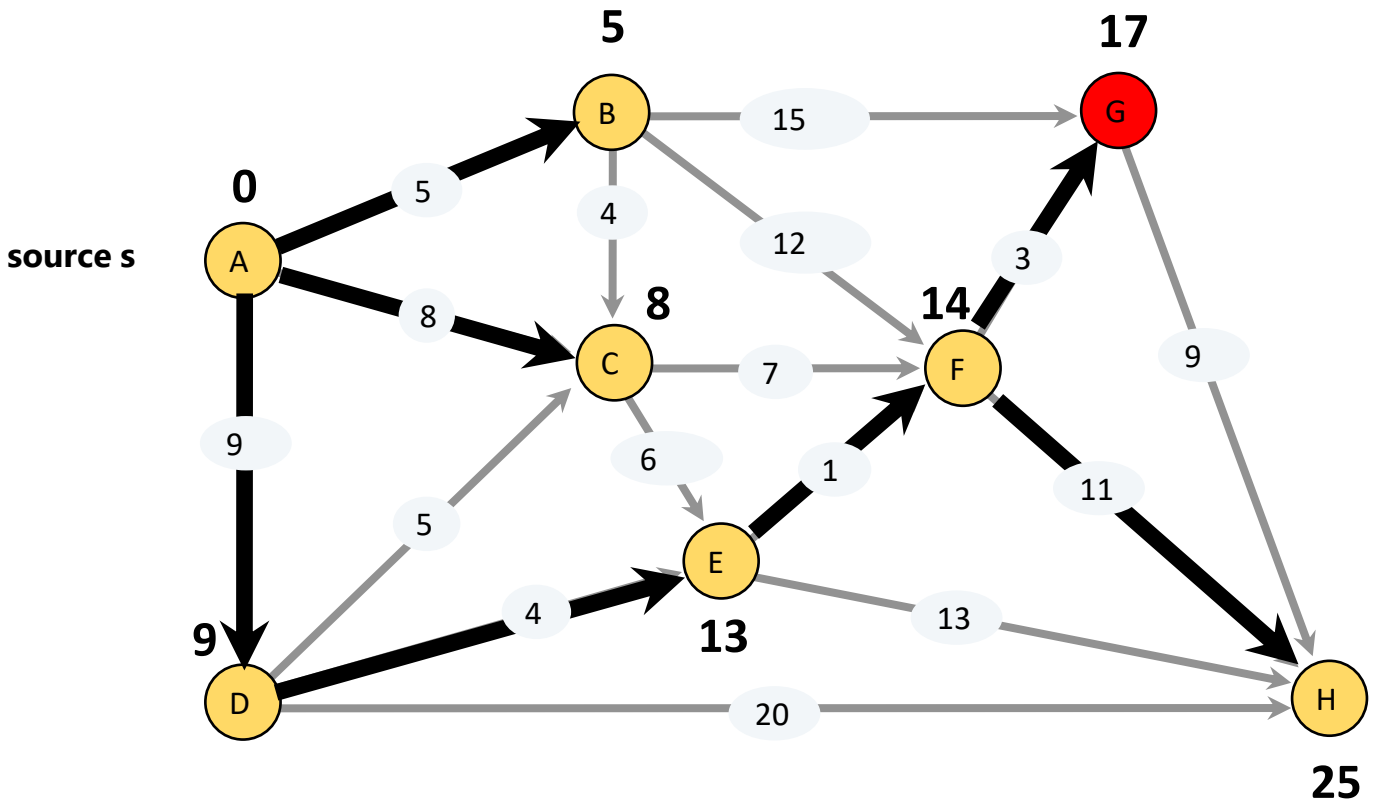
distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	17
H	25

Priority Queue

G	17
H	25

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	F
H	F

Single-source shortest paths problem



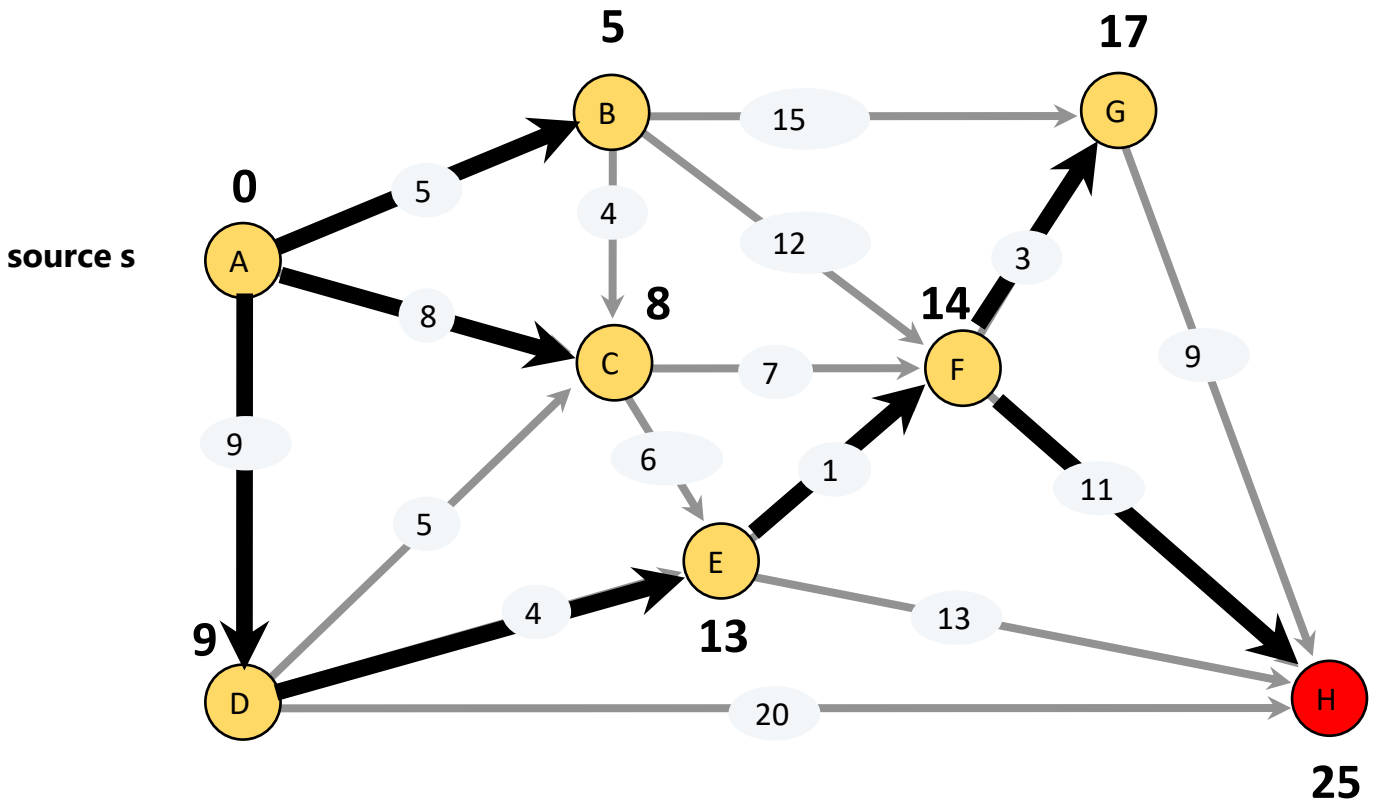
distance[ ]	
A	0
B	5
C	8
D	9
E	13
F	14
G	17
H	25

Priority Queue

H	25

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	F
H	F

Single-source shortest paths problem

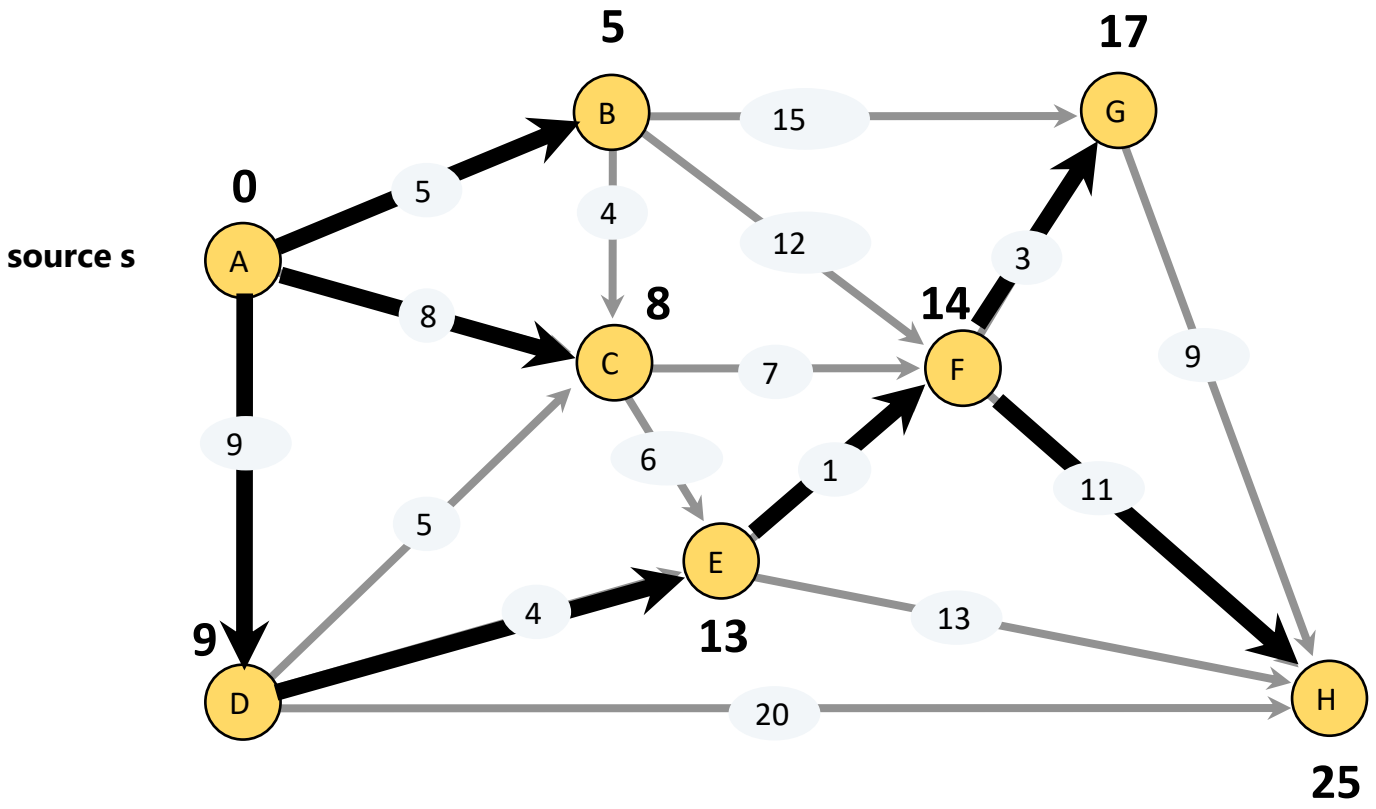


distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	17
H	25

Priority Queue

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	F
H	F

Single-source shortest paths problem



distance[ ]	
A	**0**
B	5
C	8
D	9
E	13
F	14
G	17
H	25

Priority Queue

predecessor[ ]	
A	
B	A
C	A
D	A
E	D
F	E
G	F
H	F

Dijkstra's algorithm (for single-source shortest paths problem)

```
for each vertex v in graph:
        distance[v] = ∞
        predecessor[v] = undefined
```

$\boldsymbol{O(V)}$

```
distance[s] = 0

PQ = empty priority queue
for each vertex v in graph:
        PQ.insert(v, distance[v])
```

$\boldsymbol{O(V)}$

$2O(V) + O(V)\,(\mathrm{O}(\log V)$ +out-degree(u) $\mathrm{O}(\log V))$

$2O(V) + O\,(\mathrm{V}\log V) + O(V)$(out-degree(u)) $\mathrm{O}(\log V))$

$2O(V) + O\,(\mathrm{V}\log V) + O(E)\,\mathrm{O}(\log V))$

$\boldsymbol{2O(V) + O(V + E)\,(\log V))}$

```
while PQ is not empty:
        u = PQ.extract-min()
        for each out-degree(u,v) of u:
            alt = distance[u] + weight(u, v)
            if alt < distance[v]:
                distance[v] = alt
                predecessor[v] = u
                PQ.decrease-key(v, alt)
```

$\mathbf{O}(\boldsymbol{\log V})$

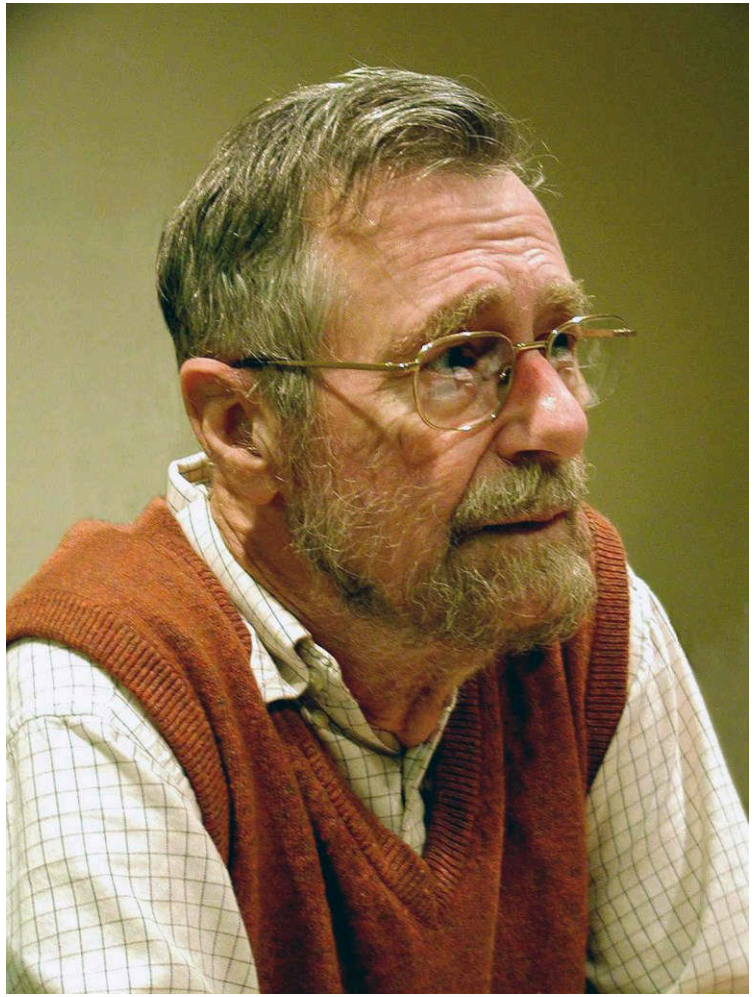
$\mathbf{O}(\boldsymbol{\log V})$

out-degree(u)

$\boldsymbol{O(V)}$

This lecture Dedicated to…

"What is the shortest way to travel from Rotterdam to Groningen? It is the algorithm for the shortest path which **I designed in about 20 minutes**. One morning I was shopping with my young fiancée, and tired, we sat down on the cafe terrace to drink a cup of coffee and I was just thinking about whether I could do this, and then designed the algorithm for the shortest path."



Edsger W. Dijkstra
Turing award 1972

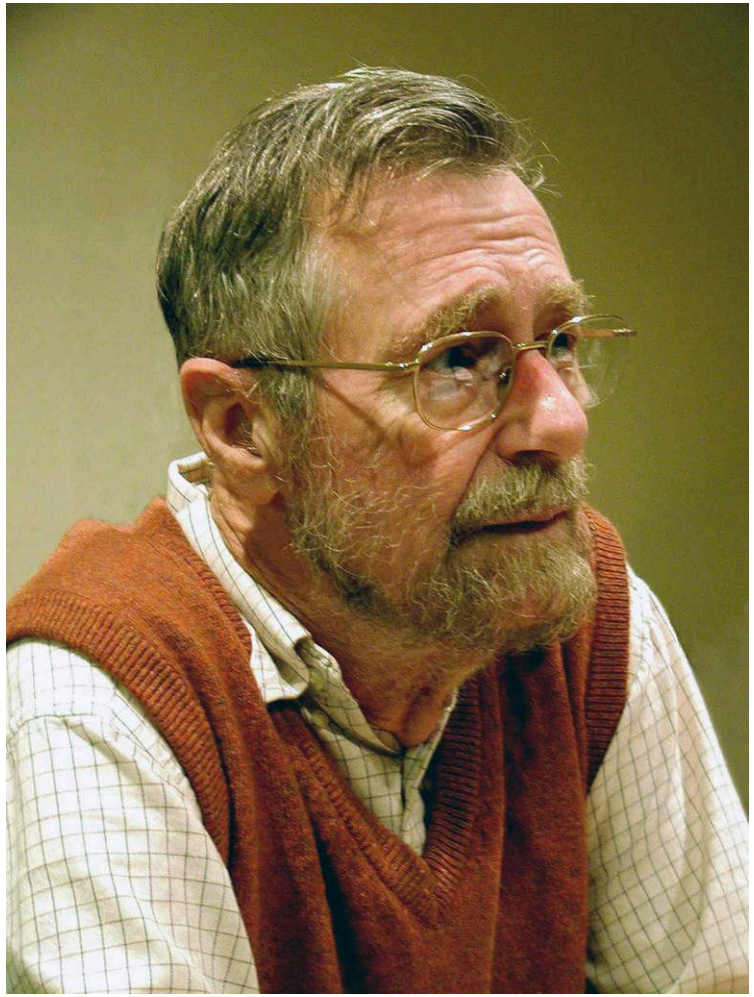
This lecture Dedicated to…

" Do only what only you can do. "

" The use of COBOL cripples the mind; its teaching should, therefore, be regarded as a criminal offence. "

" It is practically impossible to teach good programming to students that have had a prior exposure to BASIC: as potential programmers they are mentally mutilated beyond hope of regeneration. "

" APL is a mistake, carried through to perfection. It is the language of the future for the programming techniques of the past: it creates a new generation of coding bums. "



Edsger W. Dijkstra
Turing award 1972

Thanks a lot

If you are taking a Nap, **wake up**.....Lecture Over